CFA Level I Mock Exam 7.1 - Mark Scheme

Each question lists the correct choice and the original feedback for all three choices. Source excerpts accompany explanations where equations are PDF artwork.

Question 1

Regarding ethical standards, which of the following statements is least accurate?

Correct answer: B

A. Ethical conduct goes beyond what is legally required - Incorrect

Incorrect Answer Feedback: Incorrect because ethical conduct goes beyond what is legally required.

B. Good ethical judgment requires minimizing risks, excluding reputational risk - Correct

Correct Answer Feedback: Correct because good ethical judgment requires actively considering the interests of stakeholders and trying to benefit multiple stakeholders—clients, family, colleagues, employers, market participants, and so forth—and minimize risks, including reputational risk. Reputation risk is included, not excluded.

C. To act ethically, individuals need to be able to think through the facts and make good choices even in the absence of clear laws - Incorrect

Incorrect Answer Feedback: Incorrect because to act ethically, individuals need to be able to think through the facts of the situation and make good choices even in the absence of clear laws or rules.

Question 2

According to the recommended procedures for compliance with the Standard relating to additional compensation arrangements, a member should make an immediate written report specifying the proposed compensation arrangements:

Correct answer: B

A. only to her supervisor. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the recommended procedures for compliance with Standard IV(B), Additional Compensation Arrangements, members should make an immediate written report to their supervisor and compliance officer specifying any compensation they propose to receive for services.

B. only to her supervisor and to her firm’s compliance officer. - Correct

Correct Answer Feedback: Correct because according to the recommended procedures for compliance with Standard IV(B), Additional Compensation Arrangements, members and candidates should make an immediate written report to their supervisor and compliance officer specifying any compensation they propose to receive for services.

C. to her supervisor, to her firm’s compliance officer, and to her clients. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the recommended procedures for compliance with Standard IV(B), Additional Compensation Arrangements, members and candidates should make an immediate written report to their supervisor and compliance officer – and not their clients – specifying any compensation they propose to receive for services.

Question 3

According to the Standards, members are required to disclose conflicts of interest to:

Correct answer: C

A. clients only. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VI (A), Disclosure of Conflicts, disclosure is required to clients, employers and prospective clients.

B. clients and employers only. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VI (A), Disclosure of Conflicts, disclosure is required to clients, employers and prospective clients.

C. clients, employers, and prospective clients. - Correct

Correct Answer Feedback: Correct because Standard VI (A), Disclosure of Conflicts, protects investors and employers by requiring members and candidates to fully disclose to clients, potential clients, and employers all actual and potential conflicts of interest.

Question 4

A member is required to keep information confidential relating to which to the following?

Correct answer: C

A. Only prospective clients the member met within the last year - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (E), Preservation of Confidentiality, the member must keep information about current, former, and prospective clients confidential.

B. Only former clients who left the firm more than seven years ago - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (E), Preservation of Confidentiality, the member must keep information about current, former, and prospective clients confidential.

C. Both prospective clients the member met within the last year and former clients who left the firm more than seven years ago - Correct

Correct Answer Feedback: Correct because according to Standard III (E), Preservation of Confidentiality, members and candidates must keep information about current, former, and prospective clients confidential unless: 1. The information concerns illegal activities on the part of the client; 2. Disclosure is required by law; or 3. The client or prospective client permits disclosure of the information.

Question 5

Vanraj Shah has just finished taking the Level II CFA exam for the second time. To protect the integrity of the exam, Gupta is careful not to discuss the exam questions with other candidates. Shah calls his friend, a member, and contrasts his two attempts by highlighting that there were many calculation questions on derivatives in the first instance and none in the second. Later, in a public blog on investment education, Shah shares his strong disagreement regarding CFA Institute shifting to computer-based testing. Shah has violated the Standard(s) relating to:

Correct answer: A

A. conduct as participants in CFA Institute programs only. - Correct

Correct Answer Feedback: Correct because according to Standard VII(A), Conduct as Participants in CFA Institute Programs, examples of information that cannot be disclosed by candidates sitting for an exam include but are not limited to broad topical areas and formulas tested or not tested on the exam. Shah has violated Standard VII(A) by mentioning that there were no calculation questions on derivatives. Also according to Standard VII(B) relating to Reference to CFA Institute, the CFA designation, and the CFA program, when referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, Members and Candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program. Shah has not done this and hence Standard VII(B) is not violated. Only Standard VII(A) is violated.

B. reference to CFA Institute, the CFA designation, and the CFA program only. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII(B), Reference to CFA Institute, the CFA Designation, and the CFA Program, when referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, members and candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program. Shah has not done this and hence Standard VII(B) is not violated.

C. both conduct as participants in CFA Institute programs and reference to CFA Institute, the CFA designation, and the CFA program. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII(B), Reference to CFA Institute, the CFA designation, and the CFA program, when referring to CFA Institute, CFA Institute membership, the CFA designation, or candidacy in the CFA Program, members and candidates must not misrepresent or exaggerate the meaning or implications of membership in CFA Institute, holding the CFA designation, or candidacy in the CFA Program. Shah has not done this and hence Standard VII(B) is not violated.

Question 6

To be consistent with the Standard relating to communication with clients and prospective clients, a member:

Correct answer: B

A. must share in written form investment recommendations with clients. - Incorrect

Incorrect Answer Feedback: Incorrect because Standard V (B), Communication with Clients and Prospective Clients, states for purposes of Standard V(B), communication is not confined to a written report of the type traditionally generated by an analyst researching a security, company, or industry. A presentation of information can be made via any means of communication, including in- person recommendation or description, telephone conversation, media broadcast, or transmission by computer (e.g., on the internet).

B. should notify clients that additional information is available if a recommendation is shared in capsule form. - Correct

Correct Answer Feedback: Correct because Standard V (B), Communication with Clients and Prospective Clients, states, The nature of client communications is highly diverse—from one word (“buy” or “sell”) to in-depth reports of more than 100 pages. A communication may contain a general recommendation about the market, asset allocations, or classes of investments (e.g., stocks, bonds, real estate) or may relate to a specific security. If recommendations are contained in capsule form (such as a recommended stock list), members and candidates should notify clients that additional information and analyses are available from the producer of the report.

C. is permitted to share investment recommendations solely using a social media service to which most clients have access. - Incorrect

Incorrect Answer Feedback: Incorrect because Standard V (B), Communication with Clients and Prospective Clients, states Members and candidates using any social media service to communicate business information must be diligent in their efforts to avoid unintended problems because these services may not be available to all clients. When providing information to clients through new technologies, members and candidates should take reasonable steps to ensure that such delivery would treat all clients fairly and, if necessary, be considered publicly disseminated. Therefore, if only most clients, but not all clients, have access to a social media service used as the exclusive means for sharing recommendations (“solely”), not all clients are treated fairly.

Question 7

Michael Cane, CFA, is an investment manager. Cane’s proxy voting policy includes a provision that voting proxies may not be required in all instances if they do not benefit clients. The proxy voting policy is disclosed to his clients only upon request. Has Cane most likely violated the Standard relating to loyalty, prudence and care?

Correct answer: B

A. Yes, only because the voting of proxies is required in all instances - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (A), Loyalty, Prudence and Care, the member is in violation as the proxy voting policies should be disclosed to the clients not only upon request, as described in the response rationale for the correct answer.

B. Yes, only because the proxy voting policy is disclosed to his clients only upon request - Correct

Correct Answer Feedback: Correct because according to Standard III (A), Loyalty, Prudence and Care, part of a member's or candidate's duty of loyalty includes voting proxies in an informed and responsible manner. Proxies have economic value to a client, and members and candidates must ensure that they properly safeguard and maximize this value. An investment manager who fails to vote, casts a vote without considering the impact of the question, or votes blindly with management on nonroutine governance issues (e.g., a change in company capitalization) may violate this standard. Voting of proxies is an integral part of the management of investments. A cost–benefit analysis may show that voting all proxies may not benefit the client, so voting proxies may not be necessary in all instances. Members and candidates should disclose to clients their proxy voting policies. Therefore members should disclose their proxy voting policies to clients, not only upon request.

C. Yes, both because the voting of proxies is required in all instances and because the proxy voting policy is disclosed to his clients only upon request - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (A), Loyalty, Prudence and Care, voting proxies may not be necessary in all instances, as described in the response rationale for the correct answer.

Question 8

Which of the following member actions violates the Standard relating to misconduct? • Action 1: Declaring personal bankruptcy because of a large loss from investing in a startup company. • Action 2: Allocating for her personal account shares from an oversubscribed IPO that is suitable for her firm's clients

Correct answer: B

A. Action 1 only - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard I(D), Misconduct, the Code and Standards are primarily aimed at conduct and actions related to a member's or candidate's professional life. Also, personal bankruptcy may not reflect on the integrity or trustworthiness of the person declaring bankruptcy, but if the circumstances of the bankruptcy involve fraudulent or deceitful business conduct, the bankruptcy may be a violation of this standard. Therefore, a member does not violate Standard I(D), Misconduct, by declaring personal bankruptcy because of a large loss from investing in a startup company.

B. Action 2 only - Correct

Correct Answer Feedback: Correct because Standard I(D), Misconduct, addresses all conduct that reflects poorly on the professional integrity, good reputation, or competence of members and candidates. Any act that involves lying, cheating, stealing, or other dishonest conduct is a violation of this standard if the offense reflects adversely on a member's or candidate's professional activities. Accepting allocation of shares from an issuer in an oversubscribed IPO for her personal account reflects poorly on the member's professional integrity. Therefore, the member violates Standard I(D). The member's conduct also violates Standard I(B), Independence and Objectivity. One type of benefit is the allocation of shares in oversubscribed IPOs to investment managers for their personal accounts. This practice affords managers the opportunity to make quick profits that may not be available to their clients. Such a practice is prohibited under Standard I(B).

C. Both Action 1 and Action 2 - Incorrect

Incorrect Answer Feedback: Incorrect because Standard I(D), Misconduct, addresses all conduct that reflects poorly on the professional integrity, good reputation, or competence of members and candidates. Any act that involves lying, cheating, stealing, or other dishonest conduct is a violation of this standard if the offense reflects adversely on a member's or candidate's professional activities. Accepting allocation of shares from an issuer in an oversubscribed IPO for her personal accounts reflects poorly on the member's professional integrity but declaring personal bankruptcy does not relate to the member's professional life. Therefore, the former is a violation of Standard I(D), Misconduct, while the latter is not.

Question 9

Which of the following is a recommended procedure for compliance with the Standard relating to performance presentation? When presenting their firm's investment performance, members should:

Correct answer: B

A. present the performance using a single representative account. - Incorrect

Incorrect Answer Feedback: Incorrect because a composite should be used, not a single representative account. According to the Standard relating to performance presentation, members and candidates can also meet their obligations under the Standard by presenting the performance of the weighted composite of similar portfolios rather than using a single representative account.

B. maintain the data and records used to calculate the performance being presented. - Correct

Correct Answer Feedback: Correct because according to the Standard relating to performance presentation, members and candidates can also meet their obligations under the Standard by maintaining the data and records used to calculate the performance being presented.

C. exclude terminated accounts as part of performance history with a clear indication of when the accounts were terminated. - Incorrect

Incorrect Answer Feedback: Incorrect because terminated accounts should be included, not excluded. According to the Standard relating to performance presentation, members and candidates can also meet their obligations by including terminated accounts as part of performance history with a clear indication of when the accounts were terminated.

Question 10

Marcus Reid, CFA, uses the name "CFA expert" when posting comments on social media. During a conversation with a friend, Reid mentions, "I passed all three CFA exams in three consecutive years." Has Reid violated the Standards?

Correct answer: B

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII (B), Reference to the CFA institute, the CFA Designation, and the CFA Program, the member is in violation by tagging his hidden identity with the CFA designation.

B. Yes, because he uses the name "CFA expert" when posting comments on social media - Correct

Correct Answer Feedback: Correct because according to Standard VII (B), Reference to the CFA institute, the CFA Designation, and the CFA Program, the member is in violation by tagging his hidden identity with the CFA designation. Given the growing popularity of social media, where individuals may anonymously express their opinions, pseudonyms or online profile names created to hide a member's identity should not be tagged with the CFA designation.

C. Yes, because he mentions, "I passed all three CFA exams in three consecutive years" - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII (B), Reference to the CFA institute, the CFA Designation, and the CFA Program, this is a proper reference to the CFA Designation.

Question 11

Firms that claim compliance with the GIPS standards are required to:

Correct answer: A

A. include all fee-paying discretionary accounts in at least one composite. - Correct

Correct Answer Feedback: Correct because according to the GIPS standards, a claim of compliance requires that all fee-paying discretionary accounts managed by the firm be included in at least one composite.

B. determine which portfolio to include in a composite immediately after the performance reporting period has ended. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the GIPS standards, the determination of which portfolios to include in the Composite should be done according to pre-established criteria (i.e., on an ex-ante basis), not after the fact. This prevents a firm from including only their best- performing portfolios in the Composite. Therefore, it is not a requirement to determine which portfolio to include in a composite immediately after the performance reporting period has ended. Instead, firms should establish criteria beforehand, not after the fact,

C. exclude the best-performing accounts and the worst-performing accounts when representing the performance of an investment strategy. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the GIPS standards, the requirement to create, use and maintain composites is designed to prevent firms from cherry-picking—using the best-performing accounts to represent the performance of an investment strategy. However, there is no requirement to exclude best-performing and worst-performing accounts.

Question 12

Mabel Albright, CFA, resigns to work at a new firm. For her research report at the new firm, she re- creates the supporting records from her memory and uses the historical recommendations she made at her previous firm. Albright violates the Standards:

Correct answer: C

A. only by re-creating the supporting records from her memory. - Incorrect

Incorrect Answer Feedback: Incorrect because Albright violates Standard V (C), Record Retention, by performing both actions.

B. only by using the historical recommendations at her previous firm. - Incorrect

Incorrect Answer Feedback: Incorrect because Albright violates Standard V (C), Record Retention, by performing both actions.

C. both by re-creating the supporting records from her memory and by using the historical recommendations at her previous firm. - Correct

Correct Answer Feedback: Correct because according to Standard V(C), Record Retention, a member or candidate cannot use historical recommendations or research reports created at the previous firm because the supporting documentation is unavailable. For future use, the member or candidate must re-create the supporting records at the new firm with information gathered through public sources or directly from the covered company and not from memory or sources obtained at the previous employer.

Question 13

Cynthia Garcia, CFA, is an investment advisor for Synergy Investment Partners (SIP). Her new client, John Hansen, has inherited a sizeable interest in his family's business. Before reviewing his financial situation, Garcia recommends Hansen diversify his portfolio by purchasing AAA-rated long-term corporate bonds, which Garcia has recommended to several other clients. She tells Hansen "You have no downside if you hold these bonds to maturity, they are AAA-rated." Has Garcia violated the Standards?

Correct answer: C

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because Lee has violated the Standards relating to both suitability and misrepresentation as described in the correct response rationale.

B. Yes, only the Standard relating to suitability - Incorrect

Incorrect Answer Feedback: Incorrect because Lee has violated the Standards relating to both suitability and misrepresentation as described in the correct response rationale.

C. Yes, the Standard relating to suitability and the Standard relating to misrepresentation - Correct

Correct Answer Feedback: Correct because Lee has violated Standard III(C), Suitability, by not reviewing Hansen's financial situation before providing a recommendation. Standard III(C) requires that members who are in an investment advisory relationship with clients consider carefully the needs, circumstances, and objectives of the clients when determining the appropriateness and suitability of a given investment. In judging the suitability of a potential investment, the member or candidate should review many aspects of the client's knowledge, experience related to investing, and financial situation. Lee has also violated Standard I(C), Misrepresentation, by stating that Hansen cannot lose money if he holds the AAA-rated bonds to maturity. This Standard prohibits members and candidates from guaranteeing clients any specific return on volatile investments. Most investments contain some element of risk that makes their return inherently unpredictable. For such investments, guaranteeing preservation of investment capital (e.g. "I can guarantee that you will not lose money on this investment") is misleading to investors. Despite the securities being AAA-rated, a risk of default still exists (as was observed in the 2008 financial crisis).

Question 14

Thomas Turkman recently hired Georgia Viggen, CFA, as a portfolio manager for North South Bank. Although Viggen worked many years for a competitor, West Star Bank, the move was straightforward since she did not have a non-compete agreement with her previous employer. Once Viggen starts working for Turkman, the first thing she does is to bring a trading software package she developed and used at West Star to her new employer. Using public information, Viggen contacts all of her former clients to convince them to move with her to North South. Viggen also convinces one of the analysts she worked with at West Star to join her at her new employer. Viggen most likely violated the CFA Institute Code of Ethics and Standards of Professional Conduct concerning her actions involving:

Correct answer: C

A. clients. - Incorrect

Incorrect Answer Feedback: Incorrect because the member did not contact clients using information from her previous employer so she did not violate Standard IV(A)–Loyalty by these actions. Contacting former clients for any reason through the use of client lists or other information taken from a prior employer without permission would be a violation of the standard because client records are the property of the firm.

B. the analyst. - Incorrect

Incorrect Answer Feedback: Incorrect because West Star does not have any non-compete agreements so the analyst is free to seek employment elsewhere.

C. trading software. - Correct

Correct Answer Feedback: Correct because the portfolio manager violated Standard IV(A)–Loyalty by taking proprietary trading software from her former employer. Although the manager created the software, it was during a period of time when West Star employed her, so the software is not her property to take with her to her new employer. The member contacted clients using public information, so she did not violate Standard IV(A)–Loyalty. Because Viggen was not obligated to abide by a non-compete agreement that would likely restrict recruitment of former colleagues, Viggen is most likely free to recruit the analyst from his former employer.

Question 15

David Jacobs, CFA, and Megan Amari, CFA, work for Global Investment Inc. and receive fees for client referrals. Prior to signing service agreements with clients, Jacobs informs his clients about the referral fee arrangement without providing an estimated dollar value. Amari discloses all details of the arrangement, including the estimated dollar value, to her clients only after the service agreements are signed. Who has violated the Standards?

Correct answer: C

A. Amari only - Incorrect

Incorrect Answer Feedback: Incorrect as both Amari and Jacobs violated Standard VI(C), Referral Fees.

B. Jacobs only - Incorrect

Incorrect Answer Feedback: Incorrect as both Amari and Jacobs violated Standard VI(C), Referral Fees.

C. Both Amari and Jacobs - Correct

Correct Answer Feedback: Correct because according to Standards VI(C), Referral Fees, appropriate disclosure means that members and candidates must advise the client or prospective client, before entry into any formal agreement for services, of any benefit given or received for the recommendation of any services provided by the member or candidate. In addition, the member or candidate must disclose the nature of the consideration or benefit, together with the estimated dollar value. Jacobs does not provide the estimated dollar value and Amari discloses after the agreement is signed. Therefore, they both violate Standard VI(C).

Question 16

According to the GIPS standards, a firm must:

Correct answer: A

A. provide a GIPS composite report to all prospective clients. - Correct

Correct Answer Feedback: Correct because according to the GIPS standards, firms must make every reasonable effort to provide a GIPS composite report to all prospective clients when they initially become prospective clients. Firms must not choose to which prospective clients it presents a GIPS composite report.

B. include terminated composites on the firm's list of composite descriptions for at least seven years after the composite termination date. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the GIPS standards, firms must include terminated composites on this list for at least five (not seven) years after the composite termination date.

C. attain compliance for a minimum of three years or since firm inception if the firm has been in existence for less than three years to initially claim compliance. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the GIPS standards, to initially claim compliance with the GIPS standards, firms must attain compliance for a minimum of five (not three) years or for the period since the firm inception if the firm has been in existence for less than five years.

Question 17

Which of the following actions violate the Standard relating to market manipulation?

Action 1	Entering large buy and sell trades between multiple proprietary accounts of the firm with th intent to increase trading volume.	e
Action 2	Securing a dominant position in a financial instrument with an intent to influence the price of a related derivative.	
Action 3	Exploiting perceived market inefficiencies through aggressive trading strategies.	

Correct answer: A

A. Action 1 and Action 2 only - Correct

Correct Answer Feedback: Correct because Standard II(B), Market Manipulation, precludes transaction-based manipulation such as attempting to buy and sell the stock using the accounts in hopes of raising the trading volume and the price. So, Action 1 violates the Standard. Also, the Standard precludes other transaction-based manipulation such as securing a controlling, dominant position in a financial instrument to exploit and manipulate the price of a related derivative and/or the underlying asset. So, Action 2 also violates the Standard.

B. Action 1 and Action 3 only - Incorrect

Incorrect Answer Feedback: Incorrect because Standard II(B), Market Manipulation, precludes transaction-based manipulation such as attempting to buy and sell the stock using the accounts in hopes of raising the trading volume and the price. Action 1 violates the Standard. However, the Standard is not intended to preclude transactions undertaken on legitimate trading strategies based on perceived market inefficiencies. So, Action 3 does not violate the Standard.

C. Action 2 and Action 3 only - Incorrect

Incorrect Answer Feedback: Incorrect because Standard II(B), Market Manipulation, precludes other transaction-based manipulation such as securing a controlling, dominant position in a financial instrument to exploit and manipulate the price of a related derivative and/or the underlying asset. Action 2 violates the Standard. However, the Standard is not intended to preclude transactions undertaken on legitimate trading strategies based on perceived market inefficiencies. So, Action 3 does not violate the Standard.

Question 18

According to the Standards, if a member cannot discharge supervisory responsibilities due to an inadequate compliance system, the member is:

Correct answer: A

A. only required to decline in writing to accept supervisory responsibility. - Correct

Correct Answer Feedback: Correct because according to Standard IV(C), Responsibilities of Supervisor, if the member or candidate clearly cannot discharge supervisory responsibilities because of the absence of a compliance system or because of an inadequate compliance system, the member or candidate should decline in writing to accept supervisory responsibility until the firm adopts reasonable procedures to allow adequate exercise of supervisory responsibility. There is no requirement to report the firm's inadequate compliance system to the CFA Institute.

B. only required to report the inadequate compliance system to CFA Institute. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard IV(C), Responsibilities of Supervisors, there is no requirement to report the firm's inadequate compliance system to the CFA Institute.

C. both required to decline in writing to accept supervisory responsibility and to report the inadequate compliance system to CFA Institute. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard IV(C), Responsibilities of Supervisors, there is no requirement to report the firm's inadequate compliance system to the CFA Institute.

Question 19

Edo Ronde, CFA, an analyst for a hedge fund, One World Investments, is attending a key industry conference for the microelectronics industry. At lunch in a restaurant adjacent to the conference venue, Ronde sits next to a table of conference attendees and is able to read their nametags. Ronde realizes the group includes the president of a publicly traded company in the microelectronics industry, Fulda Manufacturing, a company Ronde follows. Ronde overhears the president complain about a production delay problem Fulda's factories are experiencing. The president mentions that the delay will reduce Fulda earnings more than 20% during the next year if not solved. Ronde relays this information to the portfolio manager he reports to at One World explaining that in a recent research report he recommended Fulda as a buy. The manager asks Ronde to write up a negative report on

Fulda so the fund can sell the stock. According to the CFA Institute Code of Ethics and Standards of Professional Conduct Ronde should least likely:

Correct answer: A

A. revise his research report. - Correct

Correct Answer Feedback: Correct because Ronde should refuse to follow his supervisor's request. If Ronde revises his research report based on the information he overheard at the industry conference he would violate Standard II(A), Material Nonpublic Information. The production delay information is material and considered nonpublic until it is widely distributed, so it should not be included in Ronde's research report or acted on until it becomes public. Ronde should try to encourage Fulda to make the information public.

B. leave his research report as it is. - Incorrect

Incorrect Answer Feedback: Incorrect because Ronde should do this. The production delay information is considered material and nonpublic until it is widely distributed and should not be included in Ronde's research report or acted on until it becomes public.

C. request the portfolio manager not act on the information. - Incorrect

Incorrect Answer Feedback: Incorrect because Ronde should do this. The information Ronde overheard at the industry conference is material nonpublic information and should not be acted on until it becomes public.

Question 20

To be consistent with the Standard relating to diligence and reasonable basis, members who develop quantitative models should:

Correct answer: C

A. only understand the technical aspects of the products they provide to clients. - Incorrect

Incorrect Answer Feedback: Incorrect as described in the response rationale of the correct answer.

B. only ensure the models incorporate a wide range of possible input expectations. - Incorrect

Incorrect Answer Feedback: Incorrect as described in the response rationale of the correct answer.

C. both understand the technical aspects of the products they provide to clients and ensure the models incorporate a wide range of possible input expectations. - Correct

Correct Answer Feedback: Correct because according to Standard V(A), Diligence and Reasonable Basis, members and candidates involved in the development and oversight of quantitatively oriented models, methods, and algorithms must understand the technical aspects of the products they provide to clients. Also, in reviewing the computer models or the resulting output, members and candidates need to pay particular attention to the assumptions used in the analysis and the rigor of the analysis to ensure that the model incorporates a wide range of possible input expectations, including negative market events.

Question 21

According to the Standard relating to priority of transactions, members should:

Correct answer: B

A. preclear trades in fee-paying family accounts in which the member has no beneficial ownership. - Incorrect

Incorrect Answer Feedback: Incorrect because family accounts that are client accounts should be treated like any other firm account and should neither be given special treatment nor be disadvantaged because of the family relationship. If a member or candidate has a beneficial ownership in the account, however, the member or candidate may be subject to preclearance or reporting requirements of the employer or applicable law.

B. direct their brokers to supply duplicate copies for all their personal securities transactions to their employers. - Correct

Correct Answer Feedback: Correct because according to Standard VI(B), Priority of Transactions, investment transactions for clients and employers must have priority over investment transactions in which a Member or Candidate is the beneficial owner. Also, investment personnel should be required to direct their brokers to supply to firms duplicate copies or confirmations of all their personal securities transactions and copies of periodic statements for all securities accounts.

C. allocate shares to other clients before allocating shares to fee-paying family accounts in which the members do not have beneficial ownership. - Incorrect

Incorrect Answer Feedback: Incorrect because family accounts that are client accounts should be treated like any other firm account and should neither be given special treatment nor be disadvantaged because of the family relationship. If a member or candidate has a beneficial ownership in the account, however, the member or candidate may be subject to preclearance or reporting requirements of the employer or applicable law.

Question 22

Riti Malhotra, CFA, is a research analyst covering the oil and gas sector. Malhotra visits a covered company's new oil refining facility. The company arranges a chartered flight as there are no commercial flights serving the location of the new refinery. Meals are also provided for by the company at a nearby hotel. Malhotra lets the company pay for the expenses. Has Malhotra violated the Standards?

Correct answer: A

A. No - Correct

Correct Answer Feedback: Correct because the Standards were not violated, as elaborated in the response rationales below.

B. Yes, the Standard relating to loyalty, prudence, and care - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III(A), Loyalty, Prudence and Care, members and Candidates have a duty of loyalty to their clients and must act with reasonable care and exercise prudent judgment. Members and Candidates must act for the benefit of their clients and place their clients' interests before their employer's or their own interests. Malhotra has in no way placed client interests in jeopardy and hence Standard III(A) is not violated.

C. Yes, the Standard relating to independence and objectivity - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard I(B), Independence and Objectivity, if commercial transportation is unavailable, members and candidates may accept modestly arranged travel to participate in appropriate information-gathering events, such as a property tour and the trip was strictly for business and Malhotra was not accepting irrelevant or lavish hospitality. The itinerary required chartered flights, for which analysts were not expected to pay. The accommodations were modest. These arrangements are not unusual and did not violate Standard I(B) as long as Malhotra's independence and objectivity were not compromised. Hence, Standard I(B) is not violated.

Question 23

Which of the following statements relating to the GIPS standards is accurate?

Statement 1	Verification improves the consistency of a firm's GIPS-compliant presentations.
Statement 2	Verification provides assurance to clients that specific composite presentations made by a firm are accurate.

Correct answer: A

A. Statement 1 only - Correct

Correct Answer Feedback: Correct because verification may also increase the knowledge of the firm's performance measurement team and improve the consistency and quality of the firm's GIPS standards-related performance information. Therefore, Statement 1 is accurate.

B. Statement 2 only - Incorrect

Incorrect Answer Feedback: Incorrect because verification is performed with respect to an entire firm, not on specific composites or pooled funds. Verification does not ensure the accuracy of any specific performance report. Therefore, Statement 2 is not accurate.

C. Both Statement 1 and Statement 2 - Incorrect

Incorrect Answer Feedback: Incorrect because verification is performed with respect to an entire firm, not on specific composites or pooled funds. Verification does not ensure the accuracy of any specific performance report. Therefore, Statement 2 is not accurate.

Question 24

Tim Herndon, CFA, is a senior analyst working with Brad Welch, CFA, at a large brokerage firm. The two co-author all their research reports. Herndon comes to the belief that Welch is in violation of the Standard relating to diligence and reasonable basis. Herndon discusses his concerns with Welch, his supervisor, and the firm's compliance department. All three — Welch, his supervisor, and the compliance department — dismiss his concerns. Which of the following actions should Herndon most likely take to be consistent with the Standards?

Correct answer: C

A. Threaten to leave the firm - Incorrect

Incorrect Answer Feedback: Incorrect because threatening to leave the firm will not accomplish the dissociation necessary to avoid violation of the CFA Institute Standards of Professional Conduct. In extreme cases, dissociation may require a member or candidate to actually leave his or her employment, not threaten to leave it.

B. Continue discussing the matter with Welch - Incorrect

Incorrect Answer Feedback: Incorrect because continuing to discuss the matter with Welch will not accomplish the dissociation necessary to avoid violating the CFA Institute Standards of Professional Conduct. Herndon has discussed the matter with Welch and there is no evidence further discussion will be successful.

C. Have his name removed from further research reports - Correct

Correct Answer Feedback: Correct because to avoid violating the CFA Institute Standards of Professional Conduct, Herndon should have his name removed from further research reports. If a member or candidate has reasonable grounds to believe his or her employer's activities are illegal or unethical, the member or candidate must dissociate, or separate, from the activity. Inaction combined with continuing association with those involved in the unethical conduct may be construed as participation or assistance in the unethical behavior.

Question 25

According to the GIPS standards, all fee-paying discretionary accounts must be:

Correct answer: A

A. included in at least one composite. - Correct

Correct Answer Feedback: Correct because a claim of compliance requires that all fee-paying discretionary accounts managed by the firm be included in at least one composite.

B. aggregated into composites based on size. - Incorrect

Incorrect Answer Feedback: Incorrect because a composite must include all actual, fee-paying, discretionary portfolios managed in accordance with the same investment mandate, objective, or strategy. Composites are aggregated based on investment mandate and not based on size.

C. assigned to composites on an ex-post basis. - Incorrect

Incorrect Answer Feedback: Incorrect because the determination of which portfolios to include in the Composite should be done according to pre-established criteria (i.e., on an ex-ante basis), not after the fact. The assignment cannot be on an ex post basis.

Question 26

James Fung, CFA, is the CEO of Wholecare Pharmaceutical (WPC). WPC is in its final stages of clinical trials for developing a medicine to treat heart disease. Fung posts on social media that WPC is launching a new medicine for heart disease, knowing that the news will have a positive impact on WPC's stock price. Fung has violated the Standard(s) relating:

Correct answer: C

A. only to misrepresentation. - Incorrect

Incorrect Answer Feedback: Incorrect because Fung has also violated Standard II(B), Market Manipulation. Standard II(B) requires that members and candidates uphold market integrity by prohibiting market manipulation. Market manipulation includes practices that distort security prices or trading volume with the intent to deceive people or entities that rely on information in the market. Information-based manipulation includes, but is not limited to, spreading false rumors to induce trading by others. Fung spreads the false and misleading information which pushes up WPC's stock price. Therefore, Fung has also violated Standard II(B).

B. only to market manipulation. - Incorrect

Incorrect Answer Feedback: Incorrect because Fung has also violated Standard I(C), Misrepresentation. Standard I(C) states that CFA Institute members and candidates must not knowingly make any misrepresentation relating to professional or investment activities. Fung's post was based on prediction and contrary to the facts. Fung knew or should have known that his statements were false and misleading. The online or interactive aspects of social media do not remove the need to be open and honest about the information being distributed. Fung disseminates the news of launching a new medicine for heart disease on social media. This is false and misleading information. Therefore, Fung has also violated Standard I(C).

C. both to misrepresentation and to market manipulation. - Correct

Correct Answer Feedback: Correct because Standard I(C), Misrepresentation, states that CFA Institute members and candidates must not knowingly make any misrepresentation relating to professional or investment activities. Fung's post was based on prediction and contrary to the facts. Fung knew or should have known that his statements were false and misleading. The online or interactive aspects of social media do not remove the need to be open and honest about the information being distributed. Fung disseminates the news of launching a new medicine for heart disease on social media. This is false and misleading information. Therefore, Fung has violated Standard I(C). In addition, Standard II(B), Market Manipulation, requires that members and candidates uphold market integrity by prohibiting market manipulation. Market manipulation includes practices that distort security prices or trading volume with the intent to deceive people or entities that rely on information in the market. Information-based manipulation includes, but is not limited to, spreading false rumors to induce trading by others. Fung spreads the false and misleading information which pushes up WPC's stock price. Therefore, Fung has violated Standard II(B).

Question 27

Heidi Halvorson, CFA, is the Chief Investment Officer for Tukwila Investors, an asset management firm specializing in fixed-income investments. Tukwila is in danger of losing one of its largest clients, Quinault Jewelers, which accounts for nearly one third of its revenues. Quinault recently told Halverson that Tukwila would be fired unless the performance of Quinault's portfolio improves significantly. Shortly after this conversation, Halvorson purchases two corporate bonds she believes are suitable for any of her clients based upon third party research from a reliable

and diligent source. Immediately after the purchase, one bond increases significantly in price while the other bond declines significantly. At the end of the day, Halvorson allocates the profitable bond trade to Quinault and the other bond to two of her largest institutional accounts. Halvorson most likely violated the CFA Institute Standards of Professional with regards to:

Correct answer: B

A. client suitability. - Incorrect

Incorrect Answer Feedback: Incorrect because the analyst believes the bonds are suitable for any of her clients and has not violated Standard III(C), Suitability.

B. trade allocations. - Correct

Correct Answer Feedback: Correct because the investment officer failed to deal fairly for her clients by allocating profitable trades to a favored client at the expense of others, a violation of Standard III(B), Fair Dealing. The standard requires members and candidates to treat all clients fairly when taking investment action. Tukwila should have a systematic approach to allocating trades, such as pro rata, before or at the time of trade execution or as soon as possible after trades are executed.

C. third party research. - Incorrect

Incorrect Answer Feedback: Incorrect because the analyst does have a reasonable or adequate basis for her investment decision, because it is based upon reliable third party research, and has not violated Standard V(A), Diligence and Reasonable Basis.

Question 28

If inventory is written down:

Correct answer: A

A. inventory turnover will be higher than if the write-down did not occur. - Correct

Correct Answer Feedback: Correct because inventory turnover = cost of goods sold / average inventory. An inventory write-down reduces both profit and the carrying amount of inventory on the balance sheet and thus has a negative effect on profitability, liquidity, and solvency ratios. However, activity ratios (for example, inventory turnover and total asset turnover) will be positively affected by a write-down because the asset base (denominator) is reduced.

B. the net profit margin will be higher than if the write-down did not occur. - Incorrect

Incorrect Answer Feedback: Incorrect because net profit margin = net income / revenue, which is a profitability ratio. An inventory write-down reduces both profit and the carrying amount of inventory on the balance sheet and thus has a negative effect on profitability, liquidity, and solvency ratios. As an inventory write-down reduces net profit (income) and revenues are not affected, net profit margin will be negatively affected.

C. the interest coverage ratio will be higher than if the write-down did not occur. - Incorrect

Incorrect Answer Feedback: Incorrect because interest coverage ratio = EBIT / interest payments, which is a solvency ratio. An inventory write-down reduces both profit and the carrying amount of inventory on the balance sheet and thus has a negative effect on profitability, liquidity, and solvency ratios. As an inventory write-down is recognized as an expense on the income statement, EBIT will be reduced and interest payments will not be affected. As such, interest coverage ratio will be negatively affected.

Question 29

Which of the following ratios will most likely increase as a result of an inventory write-down?

Correct answer: B

A. Current ratio - Incorrect

Incorrect Answer Feedback: Incorrect because the Current ratio would decrease instead of increase as a result of an inventory write-down. Current ratio = Current assets / Current liabilities and inventory is classified as Current assets. Among the current assets' required line items are cash and cash equivalents, trade and other receivables, inventories, and financial assets (with short maturities). Current assets would decrease while Current liabilities would remain unchanged, and the Current ratio will therefore decrease as a result of an inventory write-down.

B. Total asset turnover ratio - Correct

Correct Answer Feedback: Correct because an inventory write-down reduces both profit and the carrying amount of inventory on the balance sheet and thus has a negative effect on profitability, liquidity, and solvency ratios. However, activity ratios (for example, inventory turnover and total asset turnover) will be positively affected by a write-down because the asset base (denominator) is reduced. Total asset turnover = Revenue / Average total assets. Inventory carrying amount decreases as a result of the write-down. Thus the denominator, Average total assets decreases while the numerator, Revenue remains unchanged. Total asset turnover ratio will therefore increase as a result of an inventory write-down.

C. Receivables turnover ratio - Incorrect

Incorrect Answer Feedback: Incorrect because the Receivables turnover ratio would remain unchanged instead of increase as a result of an inventory write-down. Receivables turnover ratio = Revenue / Average receivables. Neither Revenue nor Average receivables are impacted and the Receivables turnover ratio will therefore remain unchanged as a result of an inventory write-down.

Question 30

Which of the following best describes a lin-log model?

Correct answer: A

A. Y = b + b ln X i 0 1 i - Correct

Correct Answer Feedback: Correct because in a lin-log model, the dependent variable is linear but the independent variable is logarithmic. The lin-log model takes the form of Y = b + b ln X. i 0 1 i

B. ln Y = b + b X i 0 1 i - Incorrect

Incorrect Answer Feedback: Incorrect because this describes a log-lin model, in which the dependent variable is logarithmic but the independent variable is linear.

C. ln Y = b + b ln X i 0 1 i - Incorrect

Incorrect Answer Feedback: Incorrect because this describes a log-log model, where both the dependent and independent variables are in logarithmic form.

Question 31

An analyst gathers the following information about a company:

Target D/E	0.80
Current D/E	0.65
Yield to maturity	4%
Cost of equity	9%
Tax rate	30%

The WACC to be used in financial analysis is closest to:

Correct answer: A

A. 6.24%. - Correct

Correct Answer Feedback: Correct because the target D/E ratio is used to determine WACC. The target capital structure is the company's chosen proportions of debt and equity as opposed to the actual weightings. If we know the company's target capital structure, then of course we should use this in our analysis. Weight of debt, w = (D/E)/(1+(D/E)) = 0.8/(1+0.8) = 0.4444 d Weight of equity, w = 1–w = 0.5556 e d WACC = w r (1 – t) + w r + w r =(0.4444)(4%)(1–30%)+0+(0.5556)(9%) = 6.2444% ≈ 6.24% d d p p e e where w = the target proportion of debt in the capital structure when the company raises new funds d r = the before-tax marginal cost of debt d t = the company's marginal tax rate w = the target proportion of preferred stock in the capital structure when the company raises new p funds r = the marginal cost of preferred stock p w = the target proportion of common stock in the capital structure when the company raises new e funds r = the marginal cost of common stock e

B. 6.56%. - Incorrect

Incorrect Answer Feedback: Incorrect because the current capital structure (determined by using the current D/E ratio) was used instead of the target capital structure. The target capital structure is the company's chosen proportions of debt and equity as opposed to the actual weightings. If we know the company's target capital structure, then of course we should use this in our analysis. Weight of equity, w ≠ (D/E)/(1+(D/E)) = 0.65/(1+0.65) = 0.3939 e Weight of debt, w ≠ 1–w = 0.6061 d d WACC ≠ (0.3939)(4%)(1–30%)+(0.6061)(9%) = 6.5576% ≈ 6.56%

C. 6.78%. - Incorrect

Incorrect Answer Feedback: Incorrect because the tax shield was not applied to the weighted cost of debt in the WACC formula. Weight of debt, w = (D/E)/(1+(D/E)) = 0.8/(1+0.8) = 0.4444 d Weight of equity, w = 1–w = 0.5556 e d WACC ≠ (0.4444)(4%)+(0.5556)(9%) = 6.7778% ≈ 6.78%

Question 32

The returns of a fund are as follows: Year Return (%) 1 –20.60 2 15.00 3 0.50 4 9.80 5 4.60 The mean absolute deviation (MAD) of returns for the fund is closest to:

Correct answer: A

A. 9.53%. - Correct

Correct Answer Feedback: Correct. The mean absolute deviation (MAD) for a sample is calculated as follows: where X = the return of the fund during year i i = the mean of the returns of the sample n = the number of returns in the sample i = the index for the year In this problem: Mean: = (−20.60% + 15.00% + 0.50% + 9.80% + 4.60%)/5 = 1.86% Deviations from Mean Absolute Deviation (%) |−20.60% − 1.86%| 22.46 |15.00% − 1.86%| 13.14 |0.50% − 1.86%| 1.36 |9.80% − 1.86%| 7.94 |4.60% − 1.86%| 2.74 Total 47.64 MAD = 47.64/5 9.53

B. 11.91%. - Incorrect

Incorrect Answer Feedback: Incorrect. It uses the term "n − 1" in the denominator of the formula of MAD: = 47.64%/4 = 11.91%

C. 13.69%. - Incorrect

Incorrect Answer Feedback: Incorrect. It is the sample standard deviation (SD): SD = 2 2 2 2 = {[(−20.60% − 1.86%) + (15.00% − 1.86%) + (0.50% − 1.86%) + (9.80% − 1.86%) + (4.60% − 1.86%)2]/4}0.5 = 13.69%.

Question 33

Ignoring income taxes, the recognition of an impairment loss on long-lived assets most likely results in a:

Correct answer: B

Correct Answer Feedback: Correct. The mean absolute deviation (MAD) for a sample is calculated as follows: $\mathrm{MAD} = \frac{\sum_{i=1}^{n}|X_i - \overline{X}|}{n}$ where

X_i = the return of the fund during year i

$\overline{X}$ = the mean of the returns of the sample

n = the number of returns in the sample

i = the index for the year

In this problem:

Mean: $\overline{X}$ = (−20.60% + 15.00% + 0.50% + 9.80% + 4.60%)/5 = 1.86%

Deviations from Mean	Absolute Deviation (%)
|−20.60% − 1.86%|	22.46
|15.00% − 1.86%|	13.14
|0.50% − 1.86%|	1.36
|9.80% − 1.86%|	7.94
|4.60% − 1.86%|	2.74
Total	47.64
MAD = 47.64/5	**9.53**

Figure 1: Source equation for question 32, option A

Incorrect Answer Feedback: Incorrect. It uses the term "n − 1" in the denominator of the formula of MAD:

$\frac{\sum_{i=1}^{n}|X_i - \overline{X}|}{n}$ = 47.64%/4 = 11.91%

Figure 2: Source equation for question 32, option B

Incorrect Answer Feedback: Incorrect. It is the sample standard deviation (SD):

$$\mathrm{SD} = \sqrt[2]{\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}}$$

= {[(−20.60% − 1.86%)^2 + (15.00% − 1.86%)^2 + (0.50% − 1.86%)^2 + (9.80% − 1.86%)^2 + (4.60% − 1.86%)^2]/4}$^{0.5}$ = 13.69%.

ıo answer)

Figure 3: Source equation for question 32, option C

A. lower quick ratio. - Incorrect

Incorrect Answer Feedback: Incorrect because the amount of the impairment loss will reduce the carrying amount of the asset on the balance sheet and will reduce net income on the income statement. As Quick ratio = (Cash + Short-term marketable investments + Receivables) / Current liabilities, a reduction of equipment would have no impact on the ratio.

B. higher total asset turnover ratio. - Correct

Correct Answer Feedback: Correct because the amount of the impairment loss will reduce the carrying amount of the asset on the balance sheet and will reduce net income on the income statement. Total asset turnover = Revenue / Average total assets. While Revenue (the numerator) is unaffected, the reduction of the carrying amount of the equipment (an asset) would lead to a lower average total assets; thus the ratio would increase.

C. lower cash flow from operations. - Incorrect

Incorrect Answer Feedback: Incorrect because the impairment loss is a non-cash item and will not affect cash from operations.

Question 34

In jackknife resampling:

Correct answer: A

A. the results are similar for every resample. - Correct

Correct Answer Feedback: Correct because jackknife is another resampling technique for statistical inference of population parameters. Unlike bootstrap, which repeatedly draws samples with replacement, jackknife samples are selected by taking the original observed data sample and leaving out one observation at a time from the set (and not replacing it). According to its computation procedure, we can conclude that jackknife produces similar results for every run.

B. the required number of repetitions is the same as in bootstrap resampling. - Incorrect

Incorrect Answer Feedback: Incorrect because for a sample of size n, jackknife usually requires n repetitions, whereas with bootstrap, we are left to determine how many repetitions are appropriate, i.e., the number of repetitions required is not necessarily the same as for bootstrapping.

C. each subsequent resample has one less observation than the previous resample. - Incorrect

Incorrect Answer Feedback: Incorrect because jackknife samples are selected by taking the original observed data sample and leaving out one observation at a time from the set (and not replacing it), i.e., each resample size is one less than the original data sample, not the previous resample.

Question 35

Which of the following methods results in samples that are most representative of desired subgroups in a non-homogenous population?

Correct answer: C

A. Cluster sampling - Incorrect

Incorrect Answer Feedback: Incorrect because in cluster sampling the population is divided into clusters, each of which is essentially a mini-representation of the entire populations. A major difference between cluster and stratified random samples is that in cluster sampling, a whole cluster is regarded as a sampling unit and only sampled clusters are included. In stratified random sampling, however, all the strata are included and only specific elements within each stratum are then selected as sampling unit. Also, compared with other probability sampling methods, given equal sample size, cluster sampling usually yields lower accuracy because a sample from a cluster might be less representative of the entire population.

B. Simple random sampling - Incorrect

Incorrect Answer Feedback: Incorrect because simple random sampling is particularly useful when data in the population is homogeneous and in contrast to simple random sampling, stratified random sampling guarantees that population subdivisions of interest are represented in the sample.

C. Stratified random sampling - Correct

Correct Answer Feedback: Correct because in stratified random sampling the population is divided into subpopulations (strata) based on one or more classification criteria. Simple random samples are then drawn from each stratum in sizes proportional to the relative size of each stratum in the population. This guarantees that population subdivisions of interest are represented in the sample.

Question 36

An analyst gathers the following information about three portfolios:

	Portfolio X	Portfolio Y	Portfolio Z
Expected annual return	10%	12%	14%
Standard deviation of returns	12%	16%	20%

If the shortfall level is 5%, which portfolio has the largest safety-first ratio?

Correct answer: C

A. Portfolio X - Incorrect

Incorrect Answer Feedback: Incorrect because the SF Ratio for Portfolio X is 0.417, lower than the SF Ratio of 0.450 for Portfolio Z.

B. Portfolio Y - Incorrect

Incorrect Answer Feedback: Incorrect because the SF Ratio for Portfolio Y is 0.438, lower than the SF Ratio of 0.450 for Portfolio Z.

C. Portfolio Z - Correct

Correct Answer Feedback: Correct because the safety-first ratio is calculated as SF Ratio = (E(r) – R)/σ , where E(r) is the expected return, R is the shortfall level, and σ is the standard deviation. L p L p Since the shortfall level R = 5%, the SF Ratios for the three portfolios are: L Portfolio X: (0.10 – 0.05)/0.12 = 0.417 Portfolio Y: (0.12 – 0.05)/0.16 = 0.438 Portfolio Z: (0.14 – 0.05)/0.20 = 0.450 Therefore, Portfolio Z is the best alternative according to the safety-first criterion.

Question 37

Which of the following ratios is most appropriate for evaluating a company's ability to meet short- term obligations?

Correct answer: A

A. Defensive interval - Correct

Correct Answer Feedback: Correct because liquidity ratios measure a company's ability to meet its short-term obligations, and the defensive interval ratio is a liquidity ratio that measures how long the company can continue to pay its expenses from its existing liquid assets without receiving any additional cash inflow.

B. Financial leverage - Incorrect

Incorrect Answer Feedback: Incorrect because financial leverage is a solvency ratio, and solvency ratios measure a company's ability to meet long-term obligations, not short-term obligations.

C. Working capital turnover - Incorrect

Incorrect Answer Feedback: Incorrect because working capital turnover is an activity ratio, and activity ratios measure how efficiently a company performs day-to-day tasks such as the collection of receivables and management of inventory, not its ability to meet all of its short-term obligations.

Question 38

Which of the following is a secondary source of liquidity for a company?

Correct answer: C

A. Trade credit - Incorrect

Incorrect Answer Feedback: Incorrect because this is a primary source of liquidity. Primary sources include the short-term funds, which can include items such as trade credit, bank lines of credit, and short-term investment portfolios of the firm.

B. Bank line of credit - Incorrect

Incorrect Answer Feedback: Incorrect because this is a primary source of liquidity. Primary sources include the short-term funds, which can include items such as trade credit, bank lines of credit, and short-term investment portfolios of the firm.

C. Bankruptcy protection and reorganization - Correct

Correct Answer Feedback: Correct because secondary sources used by companies include filing for bankruptcy protection and reorganization.

Question 39

An analyst gathers the following information about a company:

Cash flow from operating activities	€600,000
Preferred dividends declared and paid	€30,000
Total dividends declared and paid	€120,000
Number of common shares outstanding	100,000

The company did not issue or repurchase common shares during the year. If the company reports total dividends paid as cash flows used in operating activities, cash flow per share is:

Correct answer: C

A. €5.70. - Incorrect

Incorrect Answer Feedback: Incorrect because it calculates Cash flow per share without adding back total dividends. Accordingly, Cash flow per share = (CFO – Preferred dividends) / Number of common shares outstanding = (600 – 30) / 100= 5.7.

B. €6.30. - Incorrect

Incorrect Answer Feedback: Incorrect because it added back preferred dividends instead of common dividends. Accordingly, Cash flow per share = (CFO + Preferred dividends) / Number of common shares outstanding = (600 + 30) / 100 = 6.3.

C. €6.90. - Correct

Correct Answer Feedback: Correct because Cash flow per share = (CFO – Preferred dividends) / Number of common shares outstanding. Note that if the company reports under IFRS and includes total dividends paid as a use of cash in the operating section, total dividends should be added back to CFO as reported and then preferred dividends should be subtracted. Therefore, the calculation becomes = (CFO + Total dividends – Preferred dividends) / Number of common shares outstanding = (600 + 120 – 30) / 100 = 6.9.

Question 40

Which of the following methods is most appropriate to assess the impact of a geopolitical risk?

Correct answer: B

A. Groupthink - Incorrect

Incorrect Answer Feedback: Incorrect because teams that read similar research or speak with similar client groups may be impacted by groupthink, the practice of thinking or making decisions as a group in a way that discourages creativity or individual responsibility. For scenario analysis to be useful in portfolio management, teams must work hard to build creative processes, identify scenarios, track them, and assess the need for action on a regular cadence.

B. Scenario analysis - Correct

Correct Answer Feedback: Correct because scenario analysis is the process of evaluating portfolio outcomes across potential circumstances or states of the world. Scenarios help investment teams understand where they stand with respect to a risk that might cause them to change their behavior.

C. Linear quantitative modeling - Incorrect

Incorrect Answer Feedback: Incorrect because geopolitical risks seldom develop in linear fashion, making it difficult to monitor and forecast their likelihood, velocity, and impact on a portfolio as well as difficult to address those changes through portfolio action.

Question 41

Credit cycles:

Correct answer: A

A. can amplify business cycles. - Correct

Correct Answer Feedback: Correct because it is recognized that in a world with financial frictions, business cycles can be amplified, with deeper recessions and more extensive expansions because of changes in access to external financing. In line with this, it is found that the duration and magnitude of recessions and recoveries are often shaped by linkages between business and credit cycles.

B. are defined as fluctuations in real GDP. - Incorrect

Incorrect Answer Feedback: Incorrect because credit cycles describe the changing availability—and pricing—of credit. They describe growth in private sector credit (availability and usage of loans), which is essential for business investments and household purchases of real estate. They are therefore connected to real economic activity captured by business cycles that describe fluctuations in real GDP.

C. tend to be shorter than business cycles. - Incorrect

Incorrect Answer Feedback: Incorrect because although financial variables tend to co-vary closely with each other and can often help explain the size of an economic expansion or contraction, but they are not always synchronized with the traditional business cycle. Credit cycles tend to be longer, deeper, and sharper than business cycles.

Question 42

An analyst gathers the following information about a company:

ROE	12%
ROA	8%
Total asset turnover	2.0
Tax burden	0.4
Interest burden	0.8

Pretax margin is:

Correct answer: A

A. 10.0%. - Correct

Correct Answer Feedback: Correct because Pretax margin = EBT / Revenue; or = Interest burden × EBIT margin. Accordingly, ROA = Tax burden × Interest burden × EBIT margin × Total asset turnover; or Pretax margin = Interest burden × EBIT margin = ROA / (Total asset turnover × Tax burden); or = 8.0% / (2.0 × 0.4) = 10.0%. Pretax margin is defined as profit before tax divided by revenue.

B. 12.5%. - Incorrect

Incorrect Answer Feedback: Incorrect because it calculated EBIT margin instead of pretax margin. Accordingly, EBIT margin = ROA / (Total asset turnover × Tax burden × Interest burden); or = 8.0% / (2.0 × 0.4 × 0.8) = 12.50%.

C. 15.0%. - Incorrect

Incorrect Answer Feedback: Incorrect because it included ROE in the calculation instead of ROA. Accordingly, pretax margin = ROE / (Total asset turnover × Tax burden); or = 12.0% / (2.0 × 0.4) = 15.0%.

Question 43

An analyst gathers the following information about an investment:

Initial investment amount	$100
Value at the end of Year 1	$90
Additional amount invested at the beginning of Year 2	$10
Value at the end of Year 2	$110

The annualized time-weighted rate of return over the 2-year period is closest to:

Correct answer: A

A. −0.5%. - Correct

Correct Answer Feedback: Correct because if the investment is for more than one year, take the geometric mean of the annual returns to obtain the time-weighted rate of return over that measurement period. The two annual returns are: Year 1 return = $90/$100 – 1 = 0.9 – 1 = −0.1 and Year 2 return = $110/($90 + $10) – 1 = 1.1 – 1 = 0.1. The time-weighted rate of return is $\sqrt{(0.9 \times 1.1)} - 1 = -0.00501 \approx -0.5\%$.

B. 0.0%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the money-weighted rate of return, not the time-weighted rate of return. Calculation: $100 = –$10/(1 + r) + $110/(1 + r)2; solve for r = 0%. Calculator solution: CF = –100; CF = –10; CF = 110; compute IRR = 0%. It is also the arithmetic 0 1 2 average of annual returns, (–10% + 10%)/2 = 0%, and the result of calculating a holding period return based on the total investment over two years, $110/($100 + $10) – 1 = 0%.

C. 4.9%. - Incorrect

Incorrect Answer Feedback: Incorrect because it omits the additional investment at the beginning of Year 2. Hence, the two returns are calculated as: Year 1 return = $90/$100 – 1 = 0.9 – 1 = −0.1 and Year 2 return = $110/$90 – 1 = 1.2222 – 1 = 0.2222. The time-weighted rate of return is $\sqrt{(0.9 \times 1.2222)} - 1 = 0.0488 \approx 4.9\%$.

Question 44

If a paired comparison test of mean differences supports rejecting the null hypothesis, then the:

Correct answer: C

A. difference in means is not statistically significant. - Incorrect

Incorrect Answer Feedback: Incorrect because, if the null hypothesis is rejected, the evidence is statistically significant.

B. independence of the samples is statistically significant. - Incorrect

Incorrect Answer Feedback: Incorrect because when we want to conduct tests on two means based on samples that we believe are dependent, we use the test of the mean of the differences(a paired comparisons test). Therefore, it is not used to test whether samples are dependent or independent.

C. standard error of the mean differences is low relative to the sample mean difference. - Correct

Correct Answer Feedback: Correct because, according to the test statistic, , the lower the standard error in the denominator, the higher the value of the t-statistic. The t-statistic calculation includes the sample mean difference in the numerator. Therefore, a lower standard error (denominator) relative to the sample mean difference (numerator) results in a higher t-statistic value.

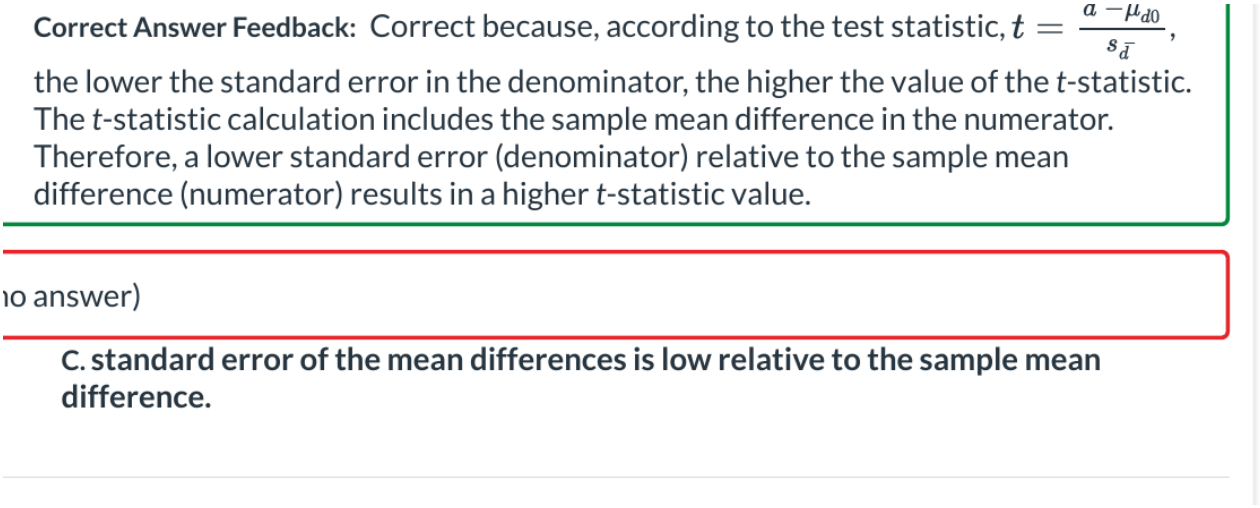


Figure 4: Source equation for question 44, option C

Question 45

Which of the following statements is most accurate? According to Modigliani–Miller Proposition I without taxes:

Correct answer: C

A. firm value can be created by changing a company's capital structure. - Incorrect

Incorrect Answer Feedback: Incorrect because MM Proposition I without taxes states: The market value of a company is not affected by the capital structure of the company. It demonstrates that managers cannot create firm value simply by changing the company's capital structure.

B. any increase in the cost of equity must exactly offset the greater use of lower cost debt. - Incorrect

Incorrect Answer Feedback: Incorrect because it is related to the Modigliani–Miller Proposition II without taxes (not Proposition I). MM Proposition II without taxes states: The cost of equity is a linear function of the company's debt-to-equity ratio. It holds that the increase in the cost of equity must exactly offset the greater use of lower cost debt.

C. equity holders demand a higher return as leverage increases in order to offset increased risk. - Correct

Correct Answer Feedback: Correct because MM Proposition I without taxes states: The market value of a company is not affected by the capital structure of the company. It demonstrates that managers cannot create firm value simply by changing the company's capital structure. In other words, the value of a company is determined solely by its cash flows, not by its relative reliance on debt and equity capital. Adding leverage increases the risk

to equity holders because greater debt increases the probability of bankruptcy. As a result, equity holders will demand a higher return as leverage increases in order to offset the increase in risk. However, overall cost of capital does not change, so there is no change in the value of the company.

Question 46

In capital investment decisions, project sequencing is most likely a type of:

Correct answer: A

A. timing option. - Correct

Correct Answer Feedback: Correct because project sequencing is a type of timing option. Delaying an investment and basing the decision on hopefully improved information that you might have in, say, a year could help improve the NPV of the projects selected. Project sequencing options allow the company to defer the decision to invest in a future investment until the outcome of some or all of a current investment is known.

B. flexibility option. - Incorrect

Incorrect Answer Feedback: Incorrect because project sequencing is a type of timing option, not a type of flexibility option. Once an investment is made, operational flexibilities besides abandonment or expansion may be available. Management may be able to exercise a price-setting option, which is a flexibility option to benefit from excess demand. There are also production flexibility options which offer the operational flexibility to alter production when demand varies from what is forecast.

C. fundamental option. - Incorrect

Incorrect Answer Feedback: Incorrect because project sequencing is a type of timing option, not a type of fundamental option. A fundamental option exists in cases where the whole investment is essentially an option. The payoffs from the investment are contingent on an underlying asset, just like most financial options.

Question 47

Under IFRS, reversals of impairments of long-lived assets are allowed:

Correct answer: A

A. for previously recognized impairment losses only. - Correct

Correct Answer Feedback: Correct. IFRS only permits the reversal of previously recognized impairment losses and does not permit the revaluation to the recoverable amount if the recoverable amount exceeds the previous carrying amount.

B. if the recoverable amount exceeds the previous carrying amount. - Incorrect

Incorrect Answer Feedback: Incorrect because IFRS do not permit the revaluation to the recoverable amount if the recoverable amount exceeds the previous carrying amount.

C. for assets held for sale only. - Incorrect

Incorrect Answer Feedback: Incorrect because IFRS allows for reversal of impairment losses regardless of whether the assets is classified as held for use or held for sale.

Question 48

For a manufacturing company reporting under US GAAP, which of the following is classified as an operating activity on the statement of cash flows?

Correct answer: C

A. Dividends paid - Incorrect

Incorrect Answer Feedback: Incorrect because under US GAAP, dividends paid is classified as a financing activity.

B. Purchase of treasury stock - Incorrect

Incorrect Answer Feedback: Incorrect because financing activities include cash payments to repurchase stock (e.g., treasury stock).

C. Purchase of trading securities - Correct

Correct Answer Feedback: Correct because additionally, operating activities include cash receipts and payments related to dealing securities or trading securities (as opposed to buying or selling securities as investments, as discussed below). In addition, the purchase and sale of securities held for dealing or trading purposes are considered operating activities even for companies where this is not a primary business activity.

Question 49

Which pricing model combines multiple products so that customers are incentivized to buy them together?

Correct answer: A

A. Bundling - Correct

Correct Answer Feedback: Correct because bundling refers to combining multiple products or services so that customers are incentivized or required to buy them together. Bundling can be effective, particularly for products that are complementary and that have high incremental profit margins and high marketing costs relative to the cost of the product itself.

B. Tiered pricing - Incorrect

Incorrect Answer Feedback: Incorrect because tiered pricing involves charging different prices to different buyers, often based on volume purchased but also based on product features (e.g., base versus premium trims of vehicles).

C. Value-based pricing - Incorrect

Incorrect Answer Feedback: Incorrect because value-based pricing means setting prices based on the value received by the customer, which often involves estimating opportunity cost.

Question 50

Which of the following strategies is least likely to be used by shareholder activists?

Correct answer: B

A. Initiating a proxy battle - Incorrect

Incorrect Answer Feedback: Incorrect because activist shareholders often pressure management through such tactics as initiating proxy battles (fights), proposing shareholder resolutions, and publicly raising awareness on issues of contention.

B. Implementing a poison pill - Correct

Correct Answer Feedback: Correct because anti-takeover measures, such as. a shareholder rights plan (also known as a poison pill), reduces the likelihood of an unwanted takeover. Shareholder rights plans enable shareholders to buy additional shares at a discount if another shareholder purchases a certain percentage of the company's shares. These plans are designed to increase the cost to any bidder seeking to take over a company. This is not a strategy used by shareholder activists but by management.

C. Filing a shareholder derivative lawsuit - Incorrect

Incorrect Answer Feedback: Incorrect because shareholder activists may pursue additional tactics, such as shareholder derivative lawsuits, which are legal proceedings initiated by one or more shareholders against board directors, management, and/or controlling shareholders of the company.

Question 51

A well-defined supply function is most likely available to:

Correct answer: B

A. a monopolist. - Incorrect

Incorrect Answer Feedback: Incorrect because as in the market structures of monopolistic competition and oligopoly, the monopolist does not have a well-defined supply function that determines the optimal output level and the price to charge.

B. a firm operating in perfect competition. - Correct

Correct Answer Feedback: Correct because in perfect competition, the firm's supply schedule is represented by the marginal cost schedule.

C. a firm operating in monopolistic competition. - Incorrect

Incorrect Answer Feedback: Incorrect because in perfect competition, the firm's supply schedule is represented by the marginal cost schedule. In monopolistic competition, there is no well-defined supply function.

Question 52

An analyst gathers the following information about three companies in the same industry but at different stages of their life cycles:

	Year 1 Revenue (in $ Millions)	Year 2 Revenue (in $ Millions)	Year 3 Revenue (in $ Millions)	Year 3 Debt/Capital
Company 1	10	11	9	0%
Company 2	30	36	44	8%
Company 3	100	95	97	25%

Based only on this information, the company with the greatest capacity to absorb additional leverage in Year 4 is most likely:

Correct answer: B

A. Company 1. - Incorrect

Incorrect Answer Feedback: Incorrect because with volatile revenue growth from a low base and no debt, this is most likely a company in its start-up phase. At this early stage, debt capital is typically not available.

B. Company 2. - Correct

Correct Answer Feedback: Correct because since the company's revenue is growing steadily (20% in the first year and 22% subsequently) and there is already a small amount of debt in the capital structure, this company is most likely in its growth phase. A company in this phase with less than 10% Debt/Capital likely has the capacity to increase debt.

C. Company 3. - Incorrect

Incorrect Answer Feedback: Incorrect because with negligible revenue growth and Debt/Capital already greater than 25%, this company is most likely a mature company. Mature companies usually use leverage, but a Debt/Capital ratio of 25% means less ability to increase leverage than a company in its growth phase with less debt outstanding.

Question 53

An analyst gathers the following exchange rate information:

USD/EUR spot rate	1.0351
1-week USD/EUR forward points	–1.1

USD/EUR is the amount of USD per 1 EUR The 1-week USD/EUR forward rate is closest to:

Correct answer: C

A. 1.02410. - Incorrect

Incorrect Answer Feedback: Incorrect because the one-week forward rate is incorrectly calculated by dividing the forward points by 100: 1.0351 + (–1.1)/100 = 1.02410.

B. 1.03400. - Incorrect

Incorrect Answer Feedback: Incorrect because the one-week forward rate is incorrectly calculated by dividing the forward points by 1,000: 1.0351 + (–1.1)/1,000 = 1.03400.

C. 1.03499. - Correct

Correct Answer Feedback: Correct because to convert any of the quoted forward points into a forward rate, one would divide the number of points by 10,000 (to scale down to the fourth decimal place, the last decimal place in the spot quote) and then add the result to the spot exchange rate quote. In this case the one-week forward rate is calculated as 1.0351 + (–1.1)/10,000 = 1.03499.

Question 54

A sign of expansionary fiscal policy is a(n):

Correct answer: B

A. increase in indirect taxes. - Incorrect

Incorrect Answer Feedback: Incorrect because an increase in indirect taxes would be contractionary not expansionary. An expansionary policy could take one or more of the following forms cuts in sales (indirect) taxes to lower prices which raises real incomes with the objective of raising consumer demand.

B. increase in the budget deficit. - Correct

Correct Answer Feedback: Correct because an increase in a budget surplus would be associated with contractionary fiscal policy, while a rise in a deficit is an expansionary fiscal policy.

C. reduction in reserve requirements for banks. - Incorrect

Incorrect Answer Feedback: Incorrect because reserve requirements are an instrument of monetary policy not fiscal policy. The third primary way in which central banks can limit or increase the supply of money in an economy is via their reserve requirements.

Question 55

An investor requiring $10,000 in ten years' time has $7,500 to invest today and has access to the following two accounts:

	Stated Annual Interest Rate	Compounding Frequency
Account 1	3.0%	Quarterly
Account 2	2.9%	Daily

Which of the accounts will allow the investor to achieve their goal?

Correct answer: C

A. Account 1 only - Incorrect

Incorrect Answer Feedback: Incorrect because Account 2 also allows the investor to meet their goal. With more than one compounding period per year, the future value formula can be expressed as FV = PV[1 + (r /m)]mN where r is the stated annual interest rate, m is the number of N s s compounding periods per year, and N now stands for the number of years. Hence for Account 2, FV 10 = $7,500 × [1 + (0.029/365)]365×10 = $10,023 which is greater than the required goal.

B. Account 2 only - Incorrect

Incorrect Answer Feedback: Incorrect because Account 1 also allows the investor to meet their goal. With more than one compounding period per year, the future value formula can be expressed as FV = PV[1 + (r /m)]mN where r is the stated annual interest rate, m is the number of N s s compounding periods per year, and N now stands for the number of years. Hence for Account 1, FV 10 4×10 = $7,500 × [1 + (0.03/4)] = $10,113 which is greater than the required goal.

C. Both Account 1 and Account 2 - Correct

Correct Answer Feedback: Correct because with more than one compounding period per year, the future value formula can be expressed as FV = PV[1 + (r /m)]mN where r is the stated annual N s s interest rate, m is the number of compounding periods per year, and N now stands for the number of years. Identifying N = 10 and PV = $7,500, we have for Account 1 (with m = 4): FV = $7,500 × [1 + 10 4×10 365×10 (0.03/4)] = $10,113 and for Account 2 (with m = 365): FV = $7,500 × [1 + (0.029/365)] 10 = $10,023. Hence, both accounts will have a balance above $10,000 in 10 years' time.

Question 56

Which of the following areas is most likely to apply natural language processing?

Correct answer: B

A. Data encryption - Incorrect

Incorrect Answer Feedback: Incorrect because data encryption uses cryptography, not natural language processing. Features of DLT (distributed ledger technology) include the use of cryptography—an algorithmic process to encrypt data, making the data unusable if received by unauthorized parties—which enables a high level of network security and database integrity. For example, DLT uses cryptographic methods of proof to verify network participant identity and for data encryption.

B. Compliance functions - Correct

Correct Answer Feedback: Correct because NLP (natural language processing) may also be employed in compliance functions to review employee voice and electronic communications for adherence to company or regulatory policy, inappropriate conduct, or fraud or for ensuring private or customer information is kept confidential. Text analytics involves the use of computer programs to analyze and derive meaning typically from large, unstructured text- or voice-based datasets, and within the larger field of text analytics, NLP is an important application.

C. Analysis of structured data - Incorrect

Incorrect Answer Feedback: Incorrect because natural language processing is applied to unstructured data, not structured data. Structured data items can be organized in tables and are commonly stored in a database where each field represents the same type of information.

Question 57

The following 10 observations are a sample drawn from a normal population: 25, 20, 18, −5, 35, 21, −11, 8, 20, and 9. The mean of the sample is closest to:

Correct answer: A

A. 14.00. - Correct

Correct Answer Feedback: Correct. The sum of the 10 numbers is 140. Dividing by 10 gives the mean of 14.

B. 15.56. - Incorrect

Incorrect Answer Feedback: Incorrect. It is calculated by dividing 140 by 9 (i.e., by n − 1 rather than n).

C. 17.20. - Incorrect

Incorrect Answer Feedback: Incorrect. It is calculated by adding the absolute values of the ten numbers (i.e., −11 is valued as 11 and −5 is valued as 5).

Question 58

The arithmetic and geometric mean are calculated for the same data. If there is variability in the data, compared with the arithmetic mean, the geometric mean will most likely be:

Correct answer: A

A. smaller. - Correct

Correct Answer Feedback: Correct. The geometric mean is always less than or equal to the arithmetic mean. The only time the two means will be equal is when there is no variability in the observations.

B. equal. - Incorrect

Incorrect Answer Feedback: Incorrect. The geometric mean is typically less than the arithmetic mean.

C. greater. - Incorrect

Incorrect Answer Feedback: Incorrect. The geometric mean is typically less than the arithmetic mean.

Question 59

A company's interest payments are fully tax deductible. When tax rates increase, the company's cost of debt:

Correct answer: A

A. decreases. - Correct

Correct Answer Feedback: Correct because the marginal cost of debt financing is the cost of debt after considering the allowable deduction for interest on debt. Based on the country's tax law the marginal cost of debt is r (1 – t). Therefore, if tax rates increase the after-tax cost of debt will d decrease.

B. remains the same. - Incorrect

Incorrect Answer Feedback: Incorrect because when interest is deductible, the marginal cost of debt is equal to r (1 – t). If tax rates increase, the cost of debt will decrease, not remain the same. d

C. increases. - Incorrect

Incorrect Answer Feedback: Incorrect because the cost of debt would decrease, not increase. The marginal cost of debt financing is the cost of debt after considering the allowable deduction for interest on debt based on the country's tax law. The marginal cost of debt is r (1 – t). d

Question 60

If a company's receivables turnover ratio is equal to its payables turnover ratio, the company's cash conversion cycle is equal to its number of days of:

Correct answer: C

A. payables. - Incorrect

Incorrect Answer Feedback: Incorrect because when a company's DSO is equal to the number of days of payables the company's cash conversion cycle is equal to its DOH. A company's cash conversion cycle is equal to DOH + DSO – Number of days of payables. The company's DSO is equal to the number of days of payables because the receivables turnover and the payables turnover ratios have the same value.

B. sales outstanding. - Incorrect

Incorrect Answer Feedback: Incorrect because when a company's DSO is equal to the number of days of payables the company's cash conversion cycle is equal to its DOH. A company's cash conversion cycle is equal to DOH + DSO – Number of days of payables. The company's DSO is equal to the number of days of payables because the receivables turnover and the payables turnover ratios have the same value.

C. inventory on hand. - Correct

Correct Answer Feedback: Correct because a company with the same value for receivables turnover and payables turnover has the same number of days sales outstanding (DSO) as the number of days of payables; therefore, the company's cash conversion cycle would equal the days of inventory on hand (DOH). That is, as a company's cash conversion cycle is equal to DOH + DSO – Number of days of payables when a company's DSO is equal to the number of days of payables, the company's cash conversion cycle is equal to its DOH.

Question 61

In a fixed parity currency regime, excess private demand for the domestic currency:

Correct answer: C

A. implies a depleting stock of foreign exchange reserves. - Incorrect

Incorrect Answer Feedback: Incorrect because in a fixed parity currency regime excess private demand for the domestic currency implies a rapidly growing stock of foreign exchange reserves. Deficient demand for the currency depletes foreign exchange reserves and exerts deflationary pressure on the economy.

B. makes no implication about the country's foreign exchange reserves. - Incorrect

Incorrect Answer Feedback: Incorrect because in a fixed parity currency regime excess private demand for the domestic currency implies a rapidly growing stock of foreign exchange reserves.

C. implies a growing stock of foreign exchange reserves. - Correct

Correct Answer Feedback: Correct because in a fixed parity currency regime excess private demand for the domestic currency implies a rapidly growing stock of foreign exchange reserves, expansion of the domestic money supply, and potentially accelerating inflation.

Question 62

In a period of rising prices and stable inventory quantities, the value of ending inventory will most closely reflect current replacement value under which of the following inventory valuation methods?

Correct answer: A

A. FIFO - Correct

Correct Answer Feedback: Correct because the value of ending inventory under FIFO reflects the costs of goods purchased more recently. Because the ending inventory reflects the cost of goods purchased more recently, it will better reflect current replacement value.

B. LIFO - Incorrect

Incorrect Answer Feedback: Incorrect because under LIFO, the oldest goods purchased remain in ending inventory. The value of ending inventory reflects the cost of older goods. Because prices are changing and the value of ending inventory reflects the cost of older goods, it is not the best reflection of current replacement value.

C. Weighted average cost - Incorrect

Incorrect Answer Feedback: Incorrect because the weighted average cost method assigns the average cost of the goods available for sale during the accounting period to the units sold as well as to the units in ending inventory. The value of ending inventory under the weighted average cost method will more closely reflect current replacement value than the LIFO method, but not as closely as the FIFO method.

Question 63

An analyst gathers the following exchange rate information:

	Spot rate	Forward rate
AUD/USD	1.3889	1.3195
AUD/EUR	1.6393	1.4875

AUD/USD is the amount of AUD per 1 USD AUD/EUR is the amount of AUD per 1 EUR The USD is expected to depreciate against:

Correct answer: A

A. the AUD only. - Correct

Correct Answer Feedback: Correct because the AUD/USD is expected to decline from 1.3889 to 1.3195, and therefore the AUD will appreciate against USD, and by definition the USD will depreciate against the AUD. The USD/EUR cross-rate is calculated as: 1/(AUD/USD) (AUD/EUR). Therefore, the spot cross rate: USD/EUR = 1/(1.3889) 1.6393 = 1.1803; and the forward cross-rate: USD/EUR = (1/1.3195) 1.4875 = 1.1273. Therefore the USD will appreciate against the EUR.

Correct Answer Feedback: Correct because the AUD/USD is expected to decline from 1.3889 to 1.3195, and therefore the AUD will appreciate against USD, and by definition the USD will depreciate against the AUD. The USD/EUR cross-rate is calculated as: 1/(AUD/USD) × (AUD/EUR). Therefore, the spot cross rate: USD/EUR = 1/(1.3889) × 1.6393 = 1.1803; and the forward cross-rate: USD/EUR = (1/1.3195) × 1.4875 = 1.1273. Therefore the USD will appreciate against the EUR.

Figure 5: Source equation for question 63, option A

B. the EUR only. - Incorrect

Incorrect Answer Feedback: Incorrect because the AUD/USD is expected to decline from 1.3889 to 1.3195, and therefore the AUD will appreciate against USD, and by definition the USD will depreciate against the AUD. The USD/EUR cross-rate is calculated as: 1/(AUD/USD) (AUD/EUR). Therefore, the spot cross rate: USD/EUR = 1/(1.3889) 1.6393 = 1.1803; and the forward cross-rate: USD/EUR = (1/1.3195) 1.4875 = 1.1273. Therefore the USD will appreciate against the EUR.

Incorrect Answer Feedback: Incorrect because the AUD/USD is expected to decline from 1.3889 to 1.3195, and therefore the AUD will appreciate against USD, and by definition the USD will depreciate against the AUD. The USD/EUR cross-rate is calculated as: 1/(AUD/USD) × (AUD/EUR). Therefore, the spot cross rate: USD/EUR = 1/(1.3889) × 1.6393 = 1.1803; and the forward cross-rate: USD/EUR = (1/1.3195) × 1.4875 = 1.1273. Therefore the USD will appreciate against the EUR.

Figure 6: Source equation for question 63, option B

C. both the AUD and the EUR. - Incorrect

Incorrect Answer Feedback: Incorrect because the AUD/USD is expected to decline from 1.3889 to 1.3195, and therefore the AUD will appreciate against USD, and by definition the USD will depreciate against the AUD. The

USD/EUR cross-rate is calculated as: 1/(AUD/USD) (AUD/EUR). Therefore, the spot cross rate: USD/EUR = 1/(1.3889) 1.6393 = 1.1803; and the forward cross- rate: USD/EUR = (1/1.3195) 1.4875 = 1.1273. Therefore the USD will appreciate against the EUR.

Question 64

All else being equal, an increase in the central bank's policy rate most likely puts downward pressure on domestic inflation by:

Correct answer: C

A. increasing consumption growth. - Incorrect

Incorrect Answer Feedback: Incorrect because market participants would then come to the view that higher interest rates will lead to slower economic growth, reduced profits, and reduced borrowing to finance asset purchases. The fall in asset prices as well as an increase in prices would reduce household financial wealth and therefore lead to a reduction in consumption growth.

B. improving investors' confidence. - Incorrect

Incorrect Answer Feedback: Incorrect because a rise in the central bank's policy rate is likely to weaken investors' confidence. Market participants would then come to the view that higher interest rates will lead to slower economic growth, reduced profits, and reduced borrowing to finance asset purchases.

C. strengthening the country's currency. - Correct

Correct Answer Feedback: Correct because there is a whole range of interconnected ways in which a rise in the central bank's policy rate can reduce real domestic demand and net external demand (that is, the difference between export and import consumption). Weaker total demand would tend to put downward pressure on the rate of domestic inflation—as would a stronger currency, which would reduce the prices of imports. Taken together, these might begin to put downward pressure on the overall measure of inflation.

Question 65

An analyst is reviewing the property, plant, and equipment disclosure related to a company's warehouse. Under the revaluation model, the analyst would least likely be able to determine:

Correct answer: A

A. the original date of acquisition. - Correct

Correct Answer Feedback: Correct because IFRS does not require disclosure of the original date of acquisition.

B. how the fair value was obtained. - Incorrect

Incorrect Answer Feedback: Incorrect because the details of how fair value was obtained must be disclosed by companies using the revaluation model.

C. the carrying amount under the cost model. - Incorrect

Incorrect Answer Feedback: Incorrect because under the cost model, the carrying amount must be disclosed by companies using the revaluation model.

Question 66

A crucial requirement for a central bank to be effective is to:

Correct answer: B

A. maintain reasonable inflation targets. - Incorrect

Incorrect Answer Feedback: Incorrect because independence, credibility, and transparency are arguably the crucial ingredients for an effective central bank, whether they target inflation or not. Thus, maintaining reasonable inflation targets (or even targeting inflation) is not a crucial requirement for a central bank to be considered effective.

B. maintain independence from the government. - Correct

Correct Answer Feedback: Correct because independence, credibility, and transparency are arguably the crucial ingredients for an effective central bank, whether they target inflation or not.

C. preserve the confidentiality of its economic views. - Incorrect

Incorrect Answer Feedback: Incorrect because independence, credibility, and transparency are arguably the crucial ingredients for an effective central bank, whether they target inflation or not. Additionally, one way of establishing credibility is for a central bank to be transparent in its decision making. By explaining their views on the economy and by being transparent in decision making, the independent, inflation-targeting central banks seek to gain reputation and credibility, making it easier to influence inflation expectations and hence ultimately easier to meet the inflation target. Thus, preserving the confidentiality of its economic views contradicts a crucial ingredient of transparency.

Question 67

In ESG factor analysis, a company's ownership of stranded assets is most likely considered a(n):

Correct answer: A

A. environmental risk. - Correct

Correct Answer Feedback: Correct because environmental factors that are generally considered material in investment analysis include the existence of carbon assets. Stranded assets are carbon- intensive assets at risk of no longer being economically viable because of changes in regulation or investor sentiment.

B. social risk. - Incorrect

Incorrect Answer Feedback: Incorrect because stranded assets are an environmental risk, not a social risk. Social factors considered in ESG implementation generally pertain to the management of the human capital of a business.

C. governance risk. - Incorrect

Incorrect Answer Feedback: Incorrect because the existence of "stranded assets" is more likely related to environmental risk, not a governance risk. Corporate governance can be defined as the system of internal controls and procedures by which individual companies are managed.

Question 68

A company's cash contribution to its defined contribution pension plan is recorded on the company's statement of cash flows as a(n):

Correct answer: A

A. operating cash outflow. - Correct

Correct Answer Feedback: Correct because under a defined-contribution plan, a company contributes an agreed-upon (defined) amount into the plan. The agreed-upon amount is the pension expense. The amount the company contributes to the plan is treated as an operating cash outflow.

B. investing cash outflow. - Incorrect

Incorrect Answer Feedback: Incorrect because the amount the company contributes to the plan is treated as an operating cash outflow, not an investing cash outflow.

C. financing cash outflow. - Incorrect

Incorrect Answer Feedback: Incorrect because the amount the company contributes to the plan is treated as an operating cash outflow, not a financing cash outflow.

Question 69

An analyst gathers the following common-size income statements for three companies in the same industry.

	Company X	Company Y	Company Z
Revenue	100%	100%	100%
Cost of goods sold	65	50	30
Administrative expenses	20	20	20
Research and development	0	5	30
Advertising expenses	5	15	10
Operating profit	10	10	10

Based on this information, which company is most likely to have a technically superior product?

Correct answer: C

A. Company X - Incorrect

Incorrect Answer Feedback: Incorrect because Company X spends nothing on research and development. It is unlikely to have a technically superior product.

B. Company Y - Incorrect

Incorrect Answer Feedback: Incorrect because Company Y spends significantly less than Company Z on research and development. It is unlikely to have a technically superior product.

C. Company Z - Correct

Correct Answer Feedback: Correct because Company Z spent the most on research and development and is able to support the highest gross margin (lowest cost of goods sold). It likely has the technically superior product.

Question 70

An analyst gathers the following information about a company's fiscal year beginning on 1 January:

Net income	£1,800,000
Preferred dividends declared and paid	£600,000
Common shares outstanding on 31 December	300,000
Weighted average common shares outstanding during the year	500,000
Convertible preferred shares outstanding during the year	25,000

If one preferred share can be converted into four common shares and there are no other potentially dilutive securities outstanding, reported diluted EPS should be:

Correct answer: A

A. £2.40. - Correct

Correct Answer Feedback: Correct because reported diluted EPS is equal to basic EPS as the convertible security is antidilutive (i.e. their inclusion would result in an EPS higher than the company's basic EPS). Under IFRS and US GAAP, antidilutive securities are not included in the calculation of reported diluted EPS. Reported diluted EPS will always be less than or equal to basic EPS. Reported diluted EPS is calculated using the if-converted method. The if-converted method is based on what EPS would have been if the convertible preferred securities had been converted at the beginning of the period. Diluted EPS using the if-converted method for preferred stock is Net

income/ (Weighted average number of shares outstanding + New common shares that would have been issued at conversion); or = £1,800,000 / [500,000 + (25,000 × 4)] = £1,800,000 / 600,000 = £3.00. Basic EPS = (Net income – Preferred dividends)/ Weighted average number of shares outstanding = (£1,800,000 – £600,000) / 500,000 = £1,200,000 / 500,000 = £2.40. As Basic EPS of £2.40 is lower than the diluted EPS of £3.00, the convertible preferred stock is antidilutive, the reported dilutive EPS should thus be equal to basic EPS of £2.40.

B. £3.00. - Incorrect

Incorrect Answer Feedback: Incorrect because it does not consider that the convertible preferred stock is antidilutive and the reported dilutive EPS equals to calculated diluted EPS of £3.00.

C. £4.00. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses the number of shares outstanding at year- end, not weighted average during the year in the formula for diluted EPS and basic EPS. Diluted EPS = Net income / (Number of common shares outstanding at year-end + New common shares that would have been issued at conversion); or £ £ £ Basic EPS = (Net income – Preferred dividends)/ Number of common shares outstanding at year- end; or £ £ £ £ As Basic EPS of £4.00 is lower than the diluted EPS of £4.50, the convertible preferred stock is antidilutive, the reported dilutive EPS should thus be equal to basic EPS of £4.00.

Incorrect Answer Feedback: Incorrect because it uses the number of shares outstanding at year-end, not weighted average during the year in the formula for diluted EPS and basic EPS.

Diluted EPS = Net income / (Number of common shares outstanding at year-end + New common shares that would have been issued at conversion);

or

$$\text{Diluted EPS} = \frac{£1{,}800{,}000}{[300{,}000+(25{,}000\times4)]} = \frac{£1{,}800{,}000}{400{,}000} = £4.50$$

Basic EPS = (Net income – Preferred dividends)/ Number of common shares outstanding at year-end;

or

$$\text{Basic EPS} = \frac{£1{,}800{,}000-£600{,}000}{300{,}000} = \frac{£}{1{,}200{,}000} = £4.00$$

As Basic EPS of £4.00 is lower than the diluted EPS of £4.50, the convertible preferred stock is antidilutive, the reported dilutive EPS should thus be equal to basic EPS of £4.00.

Figure 7: Source equation for question 70, option C

Question 71

Which of the following best represents the potential for misaligned interests between shareholders and board directors when director tenures are excessively long?

Correct answer: A

A. Entrenchment - Correct

Correct Answer Feedback: Correct because if we consider the typical elements of management compensation, we can identify common examples of misalignment or conflicts including entrenchment. When the overall level of board director or manager compensation, or tenure, is excessive, the result may lead to the avoidance of risk motivated by a vested interest in keeping one's position. In such a scenario, directors may avoid speaking out against management in the interest of shareholders or other stakeholders.

B. Empire building - Incorrect

Incorrect Answer Feedback: Incorrect because empire building refers to situations when director and management compensation are high and tied to the size of the business, it can also lead to ‘growth for growth’s sake’ in which managers are motivated to pursue acquisitions and expansion that might not increase shareholder value.

C. Excessive risk taking - Incorrect

Incorrect Answer Feedback: Incorrect because excessive risk taking stems from a compensation package relying too heavily on stock grants and options (not an excessive fixed pay) which can motivate risk-taking behavior by management, since option holders participate only in upside share price moves. Similarly, too little or no use of stock grants and options awarded to managers can lead to the opposite result. This misalignment may be at odds with the company’s value creation objective or with shareholder’s desire for higher-risk, higher-reward endeavors.

Question 72

Accounts payable are:

Correct answer: B

A. financial liabilities owed by a company through a formal loan agreement. - Incorrect

Incorrect Answer Feedback: Incorrect because notes payable (not accounts payable) are financial liabilities owed by a company through a formal loan agreement.

B. amounts a company owes its vendors for purchase of goods and services. - Correct

Correct Answer Feedback: Correct because accounts payable are amounts that a company owes its vendors for purchases of goods and services. They represent the unpaid amount as of the balance sheet date of the company’s purchases on credit.

C. reported in a different section of the balance sheet from notes payable due in one year. - Incorrect

Incorrect Answer Feedback: Incorrect because accounts payable and notes payable due in one year both are current liabilities, which are reported in the current liability section of the balance sheet.

Question 73

Under International Financial Reporting Standards (IFRS), reported operating cash flows can be increased by the classification choice made for:

Correct answer: B

A. dividends paid. - Incorrect

Incorrect Answer Feedback: Incorrect. IFRS allows classification of dividends paid as either an operating or a financing cash flow. However, when dividends paid is shown as an operating cash flow, reported operating cash flows are decreased (not increased). However, if it is classified as financing cash flows, it would not impact operating cash flows.

B. interest received. - Correct

Correct Answer Feedback: Correct. IFRS allows the classification of interest received as either an operating or a investing cash flow. When interest received is classified as operating cash flows, it would increase the reported operating cash flows. However, if it is classified as investing cash flows, it would not impact operating cash flows.

C. impairment losses on fixed assets. - Incorrect

Incorrect Answer Feedback: Incorrect. The impairment loss is a non-cash item and will not affect cash from operations.

Question 74

A construction company enters into a 2-year building contract. Payment of the consideration will be at the end of the building contract when all performance obligations are met. Costs incurred provide an appropriate measure of progress towards completing the contract. The construction company has incurred 30% of the costs during the first year. Assuming it is highly probable that revenue will not be subsequently reversed, at the end of Year 1, the construction company should most likely recognize a:

Correct answer: B

A. receivable. - Incorrect

Incorrect Answer Feedback: Incorrect because when revenue is recognized, a contract asset is presented on the balance sheet. It is only at the point when all performance obligations have been met except for payment that a receivable appears on the seller's balance sheet.

B. contract asset. - Correct

Correct Answer Feedback: Correct because when revenue is recognized, a contract asset is presented on the balance sheet. It is only at the point when all performance obligations have been met except for payment that a receivable appears on the seller's balance sheet.

C. contract liability. - Incorrect

Incorrect Answer Feedback: Incorrect because when revenue is recognized, a contract asset is presented on the balance sheet. It is only at the point when all performance obligations have been met except for payment that a receivable appears on the seller's balance sheet. If consideration is received in advance of transferring good(s) or service(s), the seller presents a contract liability.

Question 75

A company will lease machinery for five years with lease payments of $6,000 to be made at the end of each quarter, with the first payment due in three months. If the interest rate is 6% per year compounded quarterly, the present value of the lease payments is closest to:

Correct answer: B

A. $101,097. - Incorrect

Incorrect Answer Feedback: Incorrect because here the lease payments are assumed to be annual (= $6,000 × 4 = $24,000) and are then discounted at 6% per year over a 5-year period to give: PV = $24,000{[1 – 1/(1 + 0.06)5]/0.06} = $101,096.73 ≈ $101,097

B. $103,012. - Correct

Correct Answer Feedback: Correct because an ordinary annuity has equal annuity payments, with the first payment starting one period into the future. In total, the annuity makes N payments, with the first payment at t = 1 and the last at t = N. We can express the present value of an ordinary annuity as the sum of the present values of each individual annuity payment. Here, the number of quarterly lease payments is 5 × 4 = 20 and the interest rate is 6%/4 = 1.5% per quarter, implying a 20 present value PV = $6,000{[1 – 1/(1 + 0.015)]/0.015} = $103,011.83 ≈ $103,012.

C. $104,557. - Incorrect

Incorrect Answer Feedback: Incorrect because, as stated in the stem, the first payment starts at the end of the first quarter. Response C - $104,5557 – is calculated by incorrectly assuming the payments start immediately, not one period into the future.

Question 76

Which of the following most likely enforces financial reporting requirements?

Correct answer: A

A. Regulatory authorities only - Correct

Correct Answer Feedback: Correct because regulatory authorities require publicly traded issuers to prepare financial reports in accordance with specified accounting standards and other securities laws and regulations. Although technically not a regulatory authority, IOSCO regulates a significant portion of the world's financial capital markets. To ensure consistent application of international financial standards (such as the Basel Committee on Banking Supervision's standards and IFRS), it is important to have uniform regulation and enforcement across national boundaries.

B. Standard-setting bodies only - Incorrect

Incorrect Answer Feedback: Incorrect because regulatory authorities require publicly traded issuers to prepare financial reports in accordance with specified accounting standards and other securities laws and regulations.

C. Both regulatory authorities and standard-setting bodies - Incorrect

Incorrect Answer Feedback: Incorrect because regulatory authorities require publicly traded issuers to prepare financial reports in accordance with specified accounting standards and other securities laws and regulations.

Question 77

The following data apply to a firm operating in perfect competition. Quantity Total Revenue Total Cost 21 $210 $138 22 $220 $145 23 $230 $154 24 $240 $165 The firm's profit maximizing output (in units) is most likely:

Correct answer: B

A. 21. - Incorrect

Incorrect Answer Feedback: Incorrect because higher quantities deliver marginal revenue above marginal cost.

B. 23. - Correct

Correct Answer Feedback: Correct because under perfect competition, economic profits are maximized when marginal revenue equals short-run marginal cost—in this case, marginal cost crosses $10 per unit. Profits are maximized at 23 units of production because marginal cost is in excess of marginal revenue at 24 units.

C. 24. - Incorrect

Incorrect Answer Feedback: Incorrect because at 24 units, marginal costs exceed marginal revenue.

Question 78

A central bank is least likely to:

Correct answer: A

A. implement fiscal policy. - Correct

Correct Answer Feedback: Correct because while central banks are usually responsible for the operation of a country's monetary policy, they are not responsible for setting or implementing fiscal policy. Monetary policy refers to central bank activities that are directed toward influencing the quantity of money and credit in an economy. By contrast, fiscal policy refers to the government's decisions about taxation and spending.

B. oversee the payments system. - Incorrect

Incorrect Answer Feedback: Incorrect because perhaps the least appreciated role of a central bank is its role in the payments system. Central banks are usually asked to oversee, regulate, and set standards for a country's payments system.

C. manage foreign currency reserves. - Incorrect

Incorrect Answer Feedback: Incorrect because most central banks will also be responsible for managing their country's foreign currency reserves and also its gold reserves.

Question 79

The possibility of new entrants into an industry is considered in the calculation of:

Correct answer: C

A. the concentration ratio. - Incorrect

Incorrect Answer Feedback: Incorrect because just like the concentration ratio, the HHI does not take the possibility of entry into account.

B. the Herfindahl–Hirschman index. - Incorrect

Incorrect Answer Feedback: Incorrect because just like the concentration ratio, the HHI does not take the possibility of entry into account.

C. neither the concentration ratio nor the Herfindahl–Hirschman index. - Correct

Correct Answer Feedback: Correct because just like the concentration ratio, the HHI does not take the possibility of entry into account.

Question 80

Private companies most likely have:

Correct answer: A

A. less share price transparency compared to public companies. - Correct

Correct Answer Feedback: Correct because in contrast to public companies, the shares of a private company are not listed (do not trade on an exchange), so no visible company valuation or share price transparency exists.

B. similar share price transparency as public companies. - Incorrect

Incorrect Answer Feedback: Incorrect because private companies typically have less share price transparency than public companies.

C. greater share price transparency compared to public companies. - Incorrect

Incorrect Answer Feedback: Incorrect because private companies typically have less share price transparency than public companies.

Question 81

An analyst gathers the following information (in € thousands) about a company:

Purchases from suppliers	9,000
Increase in inventory	1,000
Increase in accounts payable	1,500

Cash paid to suppliers (in € thousands) is:

Correct answer: A

A. 7,500. - Correct

Correct Answer Feedback: Correct because cash paid to suppliers can be calculated by adjusting purchases for the change in accounts payable. Accordingly, Cash paid to suppliers = Purchases from suppliers – Increase in accounts payable = 9,000 – 1,500 = 7,500.

B. 8,000. - Incorrect

Incorrect Answer Feedback: Incorrect because it determines the cost of sales (instead of cash paid to suppliers) as Purchases from suppliers – Increase in inventory = 9,000 – 1,000 = 8,000.

C. 8,500. - Incorrect

Incorrect Answer Feedback: Incorrect because the Increase in inventory is added to and the Increase in accounts payable is deducted from Purchases from suppliers (instead of cost of sales) to determine cash paid to suppliers, as follows: Cash paid to suppliers = Purchases from suppliers + Increase in inventory – Increase in accounts payable = 9,000 + 1,000 – 1,500 = 8,500.

Question 82

An analyst gathers the following information about the incremental after-tax cash flows (in $ millions) for three independent investments: Year 0 Year 1 Year 2 Year 3 Year 4 Year 5 Investment 1 −80 0 0 60 60 — Investment 2 −125 35 60 80 −20 — 11 into Investment 3 −100 11 11 11 11 perpetuity If the required rate of return (r) is 12%, a company should most likely undertake which investment(s)?

Correct answer: A

A. Investment 1 only - Correct

Correct Answer Feedback: Correct because the NPV is calculated as follows: where required return (r) is 12 percent Investment 1: NPV = 1 = $0.84 Investment 2: NPV = 2 = −$1.69 Investment 3: NPV = 3 = −$8.33, where the present value of a perpetuity is with being the annual cash flow and r the discount rate At a required rate of return of 12 percent, investment 1 is the only investment with positive NPV.

Correct Answer Feedback: Correct because the NPV is calculated as follows:

$$\text{NPV} = \sum_{t=0}^{N} \frac{\text{CF}_t}{(1+r)^t}$$

where required return (r) is 12 percent

Investment 1: $\text{NPV}_1 = -80 + \frac{0}{(1.12)^1} + \frac{0}{(1.12)^2} + \frac{60}{(1.12)^3} + \frac{60}{(1.12)^4}$

= $0.84

Investment 2: $\text{NPV}_2 = -125 + \frac{35}{(1.12)^1} + \frac{60}{(1.12)^2} + \frac{80}{(1.12)^3} + \frac{-20}{(1.12)^4}$

= −$1.69

Investment 3: $\text{NPV}_3 = -100 + \frac{11}{0.12}$

= −$8.33, where the present value of a perpetuity is $\overline{CF}/r$ with $\overline{CF}$ being the annual cash flow and r the discount rate

At a required rate of return of 12 percent, investment 1 is the only investment with positive NPV.

Figure 8: Source equation for question 82, option A

B. Investment 1 and Investment 2 - Incorrect

Incorrect Answer Feedback: Incorrect because Investment 2 has a negative NPV and should be rejected.

C. Investment 2 and Investment 3 - Incorrect

Incorrect Answer Feedback: Incorrect because both investments 2 and 3 have negative NPVs and should be rejected.

Question 83

A company spends €10 million on start-up costs that are expensed for accounting purposes and fully amortized over five years using the straight-line method for tax purposes. If the start-up costs will result in future economic benefits, the tax base at the end of Year 1 is:

Correct answer: B

A. €0 million. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the carrying amount. The full amount has been expensed for financial reporting purposes leaving a carrying amount of 0.

B. €8 million. - Correct

Correct Answer Feedback: Correct because using the straight-line method, the company will deduct 1/5 for the taxes in Year 1 = 10 / 5 = 2, leaving a balance of 10 – 2 = 8. The tax base of an asset or liability is the amount at which the asset or liability is valued for tax purposes, whereas the carrying amount is the amount at which the asset or liability is recorded in the financial statements.

C. €10 million. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the total amount of the expense. This would be correct only if the company was not able to amortize the expenses over time for tax purposes and they had expected future economic benefits.

Question 84

Which of the following lists of terms is ordered from thinner to fatter tails compared with a normal distribution?

Correct answer: B

A. Mesokurtic, leptokurtic, platykurtic - Incorrect

Incorrect Answer Feedback: Incorrect because a distribution that has fatter tails than the normal distribution is referred to as leptokurtic or fat-tailed; a distribution that has thinner tails than the normal distribution is referred to as being platykurtic or thin-tailed; and a distribution similar to the normal distribution as concerns relative weight in the tails is called mesokurtic.

B. Platykurtic, mesokurtic, leptokurtic - Correct

Correct Answer Feedback: Correct because a distribution that has fatter tails than the normal distribution is referred to as leptokurtic or fat-tailed; a distribution that has thinner tails than the normal distribution is referred to as being platykurtic or thin-tailed; and a distribution similar to the normal distribution as concerns relative weight in the tails is called mesokurtic.

C. Leptokurtic, mesokurtic, platykurtic - Incorrect

Incorrect Answer Feedback: Incorrect because a distribution that has fatter tails than the normal distribution is referred to as leptokurtic or fat-tailed; a distribution that has thinner tails than the normal distribution is referred to as being platykurtic or thin-tailed; and a distribution similar to the normal distribution as concerns relative weight in the tails is called mesokurtic.

Question 85

The archetype of country behavior most likely characterized by globalization and cooperation is:

Correct answer: C

A. hegemony. - Incorrect

Incorrect Answer Feedback: Incorrect because hegemony is characterized by globalization and non-cooperation.

B. bilateralism. - Incorrect

Incorrect Answer Feedback: Incorrect because bilateralism is characterized by nationalism/anti- globalization and cooperation.

C. multilateralism. - Correct

Correct Answer Feedback: Correct because multilateralism is characterized by globalization and cooperation.

Question 86

With respect to a Porter's five forces analysis, which of the following best describes government policies?

Correct answer: C

A. They are viewed as a sixth force - Incorrect

Incorrect Answer Feedback: Incorrect because government is not best understood as a sixth force because government involvement is neither inherently good nor bad for industry profitability.

B. They are ignored because they affect all companies in the industry - Incorrect

Incorrect Answer Feedback: Incorrect because the best way to understand the influence of government on competition is to analyze how specific government policies affect the five competitive forces, rather than ignoring government policies.

C. They are analyzed to understand how they affect the five competitive forces - Correct

Correct Answer Feedback: Correct because the best way to understand the influence of government on competition is to analyze how specific government policies affect the five competitive forces.

Question 87

An analyst gathers the following information (in $ millions) about a company:

Average total assets	50
Average total liabilities	30

The financial leverage ratio is:

Correct answer: C

A. 0.6. - Incorrect

Incorrect Answer Feedback: Incorrect because Financial leverage ratio = Average total liabilities / Average total assets; or = $30 / $50 = 0.6.

B. 1.5. - Incorrect

Incorrect Answer Feedback: Incorrect because Financial leverage ratio = Average total liabilities / Average total equity; or = $30 / $20 = 1.5.

C. 2.5. - Correct

Correct Answer Feedback: Correct because Financial leverage ratio = Average total assets / Average total equity. In addition, Average total assets = Average total equity + Average total liabilities; or Average total equity = $20. Accordingly, Financial leverage ratio = $50 / $20 = 2.5.

Question 88

A primary aim of fiscal policy is to manage the economy through its influence on:

Correct answer: A

A. real GDP. - Correct

Correct Answer Feedback: Correct because a primary aim for fiscal policy is to help manage the economy through its influence on aggregate national output, that is, real GDP.

B. price stability. - Incorrect

Incorrect Answer Feedback: Incorrect because price stability is a goal of the monetary policy, not fiscal policy. While monetary policy may have different goals, there is one overarching objective that most seem to acknowledge explicitly, and that is the objective of maintaining price stability.

C. interest rates. - Incorrect

Incorrect Answer Feedback: Incorrect because it is monetary policy (not fiscal policy) that uses interest rate as a tool. The most obvious expression of a central bank's intentions and views comes via the interest rate it sets. The name of the official interest rate (or official policy rate or just policy rate) varies from central bank to central bank, but its purpose is to influence short- and long-term interest rates and ultimately real economic activity.

Question 89

An analyst gathers the following information about a company:

Before-tax cost of debt	5%
Marginal tax rate	30%
Cost of equity	8%
Target debt-to-equity ratio	50%

The company's WACC is equal to:

Correct answer: B

A. 5.75%. - Incorrect

Incorrect Answer Feedback: Incorrect because this distractor incorrectly interprets the debt-to- equity ratio as 50% weight for debt and 50% weight for equity. WACC ≠ w r (1 – t) + w r + w r = d d p p e e 0.5 × 0.05 × (1 – 0.30) + 0.5 × 0.08 = 5.75%.

B. 6.50%. - Correct

Correct Answer Feedback: Correct because WACC = w r (1 – t) + w r + w r . Given that there is d d p p e e no preferred stock in the capital structure, WACC = w r (1 – t) + w r . Also, a simple way of d d e e transforming a debt-to-equity ratio D/E into a weight—that is, D/(D + E)—is to divide D/E by 1 + D/E. As a result, WACC = (D/E)/[1 + (D/E)] × r × (1 – t) + 1/[1 + (D/E)] × r = 0.5/(1+0.5) × 0.05 × (1 – d e 0.30) + 1/(1 + 0.5) × 0.08 = 6.50%.

C. 7.00%. - Incorrect

Incorrect Answer Feedback: Incorrect because this distractor incorrectly omits the tax rate. WACC ≠ (D/E)/[1 + (D/E)] × r + 1/[1 + (D/E)] × r = 0.5/(1+0.5) × 0.05 + 1/(1 + 0.5) × 0.08 = 7.00%. d e

Question 90

If the relationship between the dependent variable and independent variable is linear, the regression residuals when plotted against the independent value should appear to:

Correct answer: B

A. be linear. - Incorrect

Incorrect Answer Feedback: Incorrect because when we look at the residuals of a model, what we would like to see is that the residuals are random. The residuals should not exhibit a pattern when plotted against the independent variable.

B. be random. - Correct

Correct Answer Feedback: Correct because when we look at the residuals of a model, what we would like to see is that the residuals are random. The residuals should not exhibit a pattern when plotted against the independent variable.

C. follow a pattern. - Incorrect

Incorrect Answer Feedback: Incorrect because when we look at the residuals of a model, what we would like to see is that the residuals are random. The residuals should not exhibit a pattern when plotted against the independent variable.